

Implicit GraphRAG: Knowledge Graph Signals Without LLM-Based Graph Construction

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Code: <https://github.com/kenstott/chonk> (MIT License)

Status: All per-type breakdowns and mean scores are confirmed post-canonical. HARE-Bench full matrix complete (§8.4), calib_v6 2×2 BM25 ablation complete (§8.7), GRB full-corpus runs (gpt-4o-mini) confirmed. **Status:** GraphRAG-Bench full-corpus runs (gpt-4o-mini) confirmed. HARE-Bench full matrix complete: 43 runs executed under corrected gold schemas (20 questions updated), revised typed scorer, and canonical NER (commit ff8f697, May 13 2026). All reported HARE-Bench scores are post-canonical.

Abstract

This paper applies architectural discipline to a layer of the enterprise AI stack that the field often treats as a black box. The RAG layer’s job in a modern agentic stack is narrow and well-defined: retrieve the right evidence for a single factual sub-query, synthesize a grounded answer from that evidence, and return both in a typed, contract-compliant form the next component can act on without re-parsing or re-grounding. Retrieval and generation are joint responsibilities of the same layer — a retrieval system that returns raw evidence and delegates synthesis to the planner has not completed its contract; it has leaked it. When that contract is enforced end-to-end — typed answers, evidence citations, delegation boundaries, deterministic interfaces between probabilistic components — the result is a system that is cheaper to build, more accurate, and more auditable than the monolithic alternative. Not despite the constraints. Because of them.

We demonstrate this on a production retrieval stack for enterprise agentic AI. The stack builds a knowledge graph entirely from NLP primitives — entity edges from NER co-occurrence, community structure from Louvain clustering — and fuses it with BM25 lexical search at query time. The full stack combines entity-ref-expansion (P1), community context injection (P2), lane-gated widened dense retrieval (P3), BM25 hybrid fusion (P4), Auto Domain Filter (ADF, P5), and Structured Response with evidence-compliance reprompting (SRR, P6). Index build time is approximately 8 minutes with zero LLM calls.

Against a vanilla RAG+rerank baseline (All=0.652 on GraphRAG-Bench), the system delivers **+0.060** (All=**0.712**), decomposed by ablation: **+0.026** from graph retrieval signals and **+0.034** from the SRR generation stack, both measured at n=4,072. An open-weight sovereign model (gpt-oss-120b, Apache 2.0) scores 0.708 on the same configuration — within the minimum detectable difference of the proprietary leader — confirming the retrieval gains transfer to sovereign deployment without modification.

The practitioner summary is a single convergent number: **~74%** single-shot accuracy on both single-domain and heterogeneous cross-domain corpora, measured at the full disambiguation stack. The single-domain figure (~72%, GraphRAG-Bench, LLM judge) and the heterogeneous figure (~74%, HARE-Bench, typed deterministic scorer) are essentially equivalent — and the HARE scorer is the stricter standard, eliminating judge tolerance for plausible-but-wrong answers. A frontier-model hallucination audit further confirms that GRB judge inflation is asymmetric: the vanilla baseline has a 10% false positive rate among its highest-confidence answers ($AC > 0.8$), while the full graph stack has 0% — meaning the measured GRB gains are conservative, not inflated. In a multi-step agent where the planner retries failed retrieval steps, the relevant metric is per-step accuracy at a targeted sub-query, not compound task success. The convergence is a key finding: heterogeneous multi-domain corpora are not an inherently harder retrieval target. They have specific, identifiable failure modes — structured identifier ambiguity, off-domain noise contamination, cross-domain entity alias fragmentation — each with a targeted solution in the stack. BM25 resolves identifier ambiguity; ADF eliminates off-domain noise at query time; breadcrumb indexing maintains structural context across source types. Without these disambiguation components, the heterogeneous penalty is real (0.552–0.679 across graph-only configurations). With them, accuracy converges with the single-domain ceiling. Enterprise agentic AI stacks typically operate over heterogeneous corpora — this convergence means practitioners do not face an accuracy tradeoff when combining corpus types within a shared retrieval layer.

The architecture supports domain composition natively: the Auto Domain Filter (ADF) classifies each query into one or more of its corpus’s named domains at query time and scopes retrieval accordingly, with zero index cost. A human operator’s explicit domain specification can substitute for or override ADF when available. Both paths propagate through all retrieval layers and lift effective operating accuracy toward the single-domain ceiling.

HARE-Bench, a cross-domain benchmark spanning financial filings, security advisories, regulatory documents, and technical patents, is the primary evaluation surface. GraphRAG-Bench (Medical + Novel) serves as a credibility bridge to published baselines. A 2×2 BM25 ablation (entity lane $\pm$ BM25) quantifies the contribution of lexical search relative to the practitioner’s default upgrade path and relative to graph-structural signals.

1. Introduction

The field’s dominant approach to retrieval-augmented generation has been architecturally undisciplined: a probabilistic component owns both the reasoning and the interface, producing prose that the next consumer must re-parse, re-evaluate, and often re-query to extract an actionable value. This is not a criticism of LLMs. It is a criticism of the architecture. A component that does not present a typed interface to its consumer is not a component — it is a black box that happens to be adjacent to other black boxes.

You cannot test it in isolation, audit its outputs reliably, or fix the part that is broken without touching everything downstream.

This paper demonstrates what happens when architectural discipline is applied to the retrieval layer: delegation with contracts, typed boundaries, probabilistic reasoning where generative flexibility adds value, deterministic output where the next component needs to act. The result is a system that is cheaper, more accurate, and more trustworthy — not despite those constraints, but because of them.

The benchmark name is not accidental. You’ve heard the fable of the tortoise and the hare. The hare is fast, confident, and has been sprinting hard — LLM-based graph construction, relation extraction at $O(\text{docs} \times \text{LLM-calls})$, index builds measured in hours. The tortoise reads the engineering specs: documents have structure, entities co-occur, communities form, networks carry signal. It builds the same graph in eight minutes with zero LLM calls. The tortoise wins. It wins on a standard homogeneous benchmark (GraphRAG-Bench), though the margin is narrow enough that a skeptic might attribute it to noise. It wins more clearly on HARE-Bench — a heterogeneous cross-domain benchmark designed for the conditions that actually separate retrieval strategies: incompatible entity vocabularies, cross-source evidence assembly, typed deterministic scoring that punishes plausible-but-wrong answers. GraphRAG-Bench is the flat track. HARE-Bench is the terrain. On terrain, the gap opens up and the structural approach’s advantage becomes unambiguous.

The immediate motivation for this work was a multi-step reasoning agent operating across heterogeneous corpora — relational databases and unstructured documents side by side. The agent could answer complex questions, but it did so inefficiently: through repeated re-prompting, intermediate validation steps, and iterative proof-building to assemble evidence scattered across sources. It worked. It was expensive.

Each re-prompting cycle was the agent doing manually what a better retrieval layer should have done automatically: traversing the connections between entities, following topic threads across documents, and expanding the evidence set until it was complete. The agent was compensating for a RAG layer that returned locally relevant but globally incomplete context.

This paper asks two questions. First: whether the retrieval layer itself can encode the relational structure that agents currently rediscover through re-prompting — without the cost of LLM-based graph construction. Second: whether the generator can be required to engage with the evidence it receives — without reranking infrastructure or model scale. The answer to both is yes, and the gains are measured by ablation: +0.026 from graph signals, +0.034 from the structured generation stack, measured at full-corpus scale ($n=4,072$).

Almost all real-world documents have structure. Paragraphs group related sentences. Sections group related paragraphs. Named entities recur across sections, connecting ideas that naive chunking severs. A fixed-size chunker discards this structure by design: it splits at token boundaries, not semantic ones, producing chunks whose boundaries are arbitrary with respect to the content they contain. The graph signals we recover — co-occurrence edges, community partitions, entity-ref-expansion — are not additions to

the retrieval pipeline. They are recoveries of structure the chunker discarded. The cost of recovering them via NLP primitives is negligible precisely because the structure was always there; we are reading it, not constructing it.

Existing GraphRAG systems (G-reasoner, AutoPrunedRetriever, HippoRAG2) encode this structure through LLM-based entity and relation extraction, requiring $O(\text{docs} \times \text{LLM-calls})$ at index time. For an agent operating across heterogeneous and frequently-updated corpora, that cost is impractical.

We build the graph differently. At index time, spaCy NER identifies entities in each chunk; co-occurrence edges connect entities that appear together; Louvain clustering over the co-occurrence matrix partitions the graph into communities. The result is a full knowledge graph — nodes, edges, community structure — built in minutes with no LLM calls. At query time, the graph is traversed: entity-ref-expansion follows entity edges to retrieve non-adjacent chunks, community context injection provides global topic framing, and widened dense retrieval (k gated by lane similarity threshold) approximates multi-hop path traversal.

1.1 Design Principles

Three structural signals drive the system:

P1 — Entity connectivity. NER identifies entities in each chunk. Co-occurrence within a chunk creates an edge between those entities in the graph. At query time, entity-ref-expansion follows these edges: chunks that share a named entity with a retrieved chunk are pulled in, even if they are embedding-distant. Lane filtering ($\text{sim} \geq 0.60$) gates expansion on query relevance — expansion without filtering hurts (-0.017), filtered expansion helps ($+0.017$). The lane threshold acts as the quality gate on retrieval depth: k is set large; the similarity threshold controls what lands.

P2 — Community structure. Louvain clustering over the co-occurrence matrix partitions the entity graph into topically coherent communities at $O(\text{chunks}^2)$, one-time. At query time, the communities of retrieved chunks are identified and a community context summary is prepended to the generator prompt, providing global topic framing that individual chunk retrieval misses.

P3 — Traversal depth. Wider dense retrieval ($k=30$, gated by lane sim) approximates multi-hop graph traversal by expanding the evidence pool. Redundancy pruning ($\text{cosine} \geq 0.92$) prevents context dilution as k grows.

P4 — BM25 hybrid fusion. BM25 lexical search runs in parallel with dense vector retrieval. Reciprocal Rank Fusion (RRF) merges the two ranked lists before the lane gate is applied. BM25 is particularly effective on HARE-Bench, where structured identifiers (patent numbers, CVE codes) are semantically opaque but lexically exact-matchable. Vector retrieval over a corpus of 25 Apple/Google patents cannot distinguish US12332430 from US12334096; BM25 retrieves the correct chunk in one pass.

P5 — Auto Domain Filter (ADF) — the domain routing component. For each in-

coming query, a lightweight LLM call classifies it into one or more of the corpus’s named domains (e.g., `sec`, `cve`, `patents`, `gleif`). Retrieval is then scoped to those domain namespaces, preventing off-domain noise from contaminating the candidate pool. ADF is zero-cost at index time; it adds one fast classification call at query time. ADF presupposes a corpus that has been architected and curated by domain — documents must be assigned to named namespaces at ingest. This is a deliberate design constraint: the namespace structure is the contract that makes domain routing possible, and maintaining it is part of corpus governance rather than a retrieval concern. On HARE-Bench — where every question spans a heterogeneous corpus and off-domain chunks regularly surface as top-k candidates — ADF delivers a consistent lift over the no-ADF baseline. The no-ADF BC+BM25+SRR run serves as the ADF ablation, quantifying the domain-routing contribution in isolation.

P6 — Evidence-compliance reprompt (SRR). The generator is required to produce `{answer, key_claims, evidence_used}` JSON. If `evidence_used` is empty on the first response, a single targeted reprompt enumerates the key claims and requests verbatim quotes. No additional retrieval occurs; the reprompt fires at most once and is a no-op when the generator already cited evidence. SRR is strictly additive to the retrieval stack by construction. Its gains are intuitive but model-capability-gated: compliance with the JSON schema and reprompt quality both depend on the generator’s instruction-following ability (H2, §7.3).

Because P1 operates at retrieval time, P2 at prompt-construction time, P3 at the retrieval pool level, P4 at the fusion stage, P5 at the query-routing stage, and P6 at generation time, their failure modes are orthogonal — gains compound superadditively.

1.2 Contributions

1. We build a GraphRAG system entirely from NLP primitives (NER + Louvain) and show it delivers **+0.060** over vanilla RAG+rerank ($0.652 \rightarrow \mathbf{0.712}$, $n=4,072$) at zero index-time LLM cost and an ~8-minute index build. An open-weight sovereign model (gpt-oss-120b) scores 0.708 on the same configuration — statistically tied with the proprietary leader — confirming the gains transfer to air-gapped deployment without modification (§4–7).
2. We demonstrate conditional superadditivity: entity-ref-expansion (P1), community context (P2), and widened retrieval (P3) correct orthogonal failure modes; gains compound when k is large enough that added context is non-redundant. Superadditivity is present at $k=15/20$ +pruning but not $k=10$. Redundancy pruning shifts the k -plateau, providing a principled approach to retrieval depth (§5–6).
3. We introduce Evidence-Compliance Reprompting (SRR): a generation-side feature that requires `{answer, key_claims, evidence_used}` JSON and reprompts once when no evidence is cited. SRR is strictly additive to the retrieval stack and model-capability-gated: gains are bounded by the generator’s instruction-following fidelity (§7).
4. We introduce Auto Domain Filter (ADF, P5): a query-time domain routing signal that classifies each query into the corpus’s named domains and scopes retrieval accordingly. ADF is zero-cost at index time and eliminates off-domain

noise — the primary contamination source in heterogeneous corpora. The no-ADF BC+BM25+SRR configuration is the ADF ablation; its separation from the ADF run quantifies the domain-routing contribution in isolation (§8.8).

5. We introduce HARE-Bench, a cross-domain benchmark spanning financial filings, CVE records, regulatory documents, and patents. Question design mirrors targeted sub-queries a multi-step planner would issue, making single-shot evaluation a realistic proxy for the retrieval layer in production (§8–9).
6. We establish a two-number practitioner summary — ~72% measured on GraphRAG-Bench (single-domain), projecting toward ~80% as the single-domain ceiling, and ~74% on heterogeneous cross-domain corpora — both on a typed deterministic scorer. The heterogeneous figure is discriminative by design: a benchmark where every strategy scores 80% measures nothing. Domain composition guidance (namespace filters at query time) lifts the effective per-step accuracy toward the single-domain ceiling for human-in-the-loop agents (§9).

2. Related Work

The retrieval augmentation literature divides along two axes: how the index encodes document structure, and how the retrieval pass exploits it. Systems at one extreme build no structure at all — dense retrieval over flat chunk embeddings — and compensate through better generators or rerankers. Systems at the other extreme build explicit knowledge graphs through LLM-based entity extraction and relation classification, encoding rich relational structure at high index cost. This work occupies the gap: graph structure built entirely from NLP primitives, retrieval quality approaching LLM-based systems, index cost approaching flat retrieval.

LLM-based GraphRAG systems (Microsoft GraphRAG, LightRAG, Fast-GraphRAG, HippoRAG2, G-reasoner) extract entity nodes and relation edges through LLM calls at index time. HippoRAG2 uses proposition-level chunking and LLM-generated summaries to build a hippocampus-inspired associative graph; G-reasoner adds a reasoning layer that traverses the KG at query time to resolve multi-hop questions. These systems score well on global sensemaking benchmarks — tasks requiring thematic synthesis across a corpus — precisely because the LLM-extracted relations encode semantic meaning that NER co-occurrence misses. The cost is index build time measured in hours and the requirement to rebuild on corpus updates. For continuously-updated enterprise corpora, this cost is prohibitive.

Dense retrieval with structural augmentation approaches — RAPTOR, HyDE, FLARE — improve retrieval without explicit graph construction. RAPTOR builds a recursive summary tree, enabling retrieval at multiple abstraction levels; HyDE generates hypothetical documents to bridge query-document embedding space gaps; FLARE uses active retrieval with iterative query refinement. These approaches are compatible with our system and serve as our non-graph baselines. Their common limitation is that they operate on chunks as atomic units — structural connections between chunks (entity co-occurrence, community membership) are not represented.

Lexical-semantic hybrid retrieval — BM25 fused with dense vectors via Reciprocal Rank Fusion — is the de facto production standard. Elasticsearch, OpenSearch, and Weaviate all expose hybrid search natively. The question is not whether hybrid retrieval improves over pure dense retrieval (it does, consistently, on retrieval benchmarks), but whether it substitutes for or complements graph-structural signals. This paper’s 2×2 ablation (§8.7) provides a direct answer: the BM25 contribution is isolated from the entity lane contribution, and their interaction measured. The result positions NLP-primitive graph retrieval relative to the practitioner’s default upgrade path (add BM25) rather than only relative to academic baselines.

2.1 LLM-Based GraphRAG Systems

LLM-based GraphRAG systems (LightRAG, Fast-GraphRAG, HippoRAG2, G-reasoner) construct knowledge graphs through entity recognition and relation extraction at $O(\text{docs} \times \text{LLM-calls})$ index time. These systems achieve strong benchmark results but impose hours of index build time and require full reconstruction on corpus updates — impractical for enterprise deployments where corpora churn continuously.

The modern enterprise AI stack compounds this mismatch. Real deployments do not operate on static corpora: CVE feeds update daily, 10-K filings arrive quarterly, regulatory guidance is amended continuously. LLM-based graph construction at $O(\text{docs} \times \text{LLM-calls})$ requires full reconstruction on each update — impractical at corpus churn rates that make a static index obsolete within days. The agent layer above retrieval further sharpens the requirement: planners issue targeted sub-queries whose answers must be precisely grounded, not thematically summarized. A retrieval layer that cannot return the right chunk for a specific factual sub-query forces the agent to compensate through re-prompting and iterative validation — the exact inefficiency this work sets out to eliminate.

2.2 NLP-Primitive GraphRAG (This Work)

We construct the same graph structures — entity nodes, co-occurrence edges, community partitions — using spaCy NER and Louvain clustering. Index time drops to minutes with zero LLM calls. Query-time traversal follows the same pattern: entity-ref-expansion walks entity edges; community context injection uses community membership; widened retrieval (k) extends traversal depth.

Our system uses semantic boundary chunking (1,100–2,200 characters, approximately 300–600 tokens), splitting only at natural linguistic boundaries rather than fixed token counts. To isolate the contribution of graph signals from chunking strategy, we evaluate `graph_first` retrieval on both semantic chunks and naive 256-token chunks, holding the retrieval mechanism constant.

2.3 Benchmark Scenario Selection

Published RAG benchmark scores are only meaningful relative to the task the benchmark was designed to test. Microsoft’s original GraphRAG evaluation — and the broader class of benchmarks it influenced — tests “global sensemaking”: thematic comprehension of narrative corpora rated on coverage and diversity of generated summaries. The task is explicitly thematic: “What are the main themes of this corpus?” rather than “What does this filing say about counterparty exposure as of Q3 2024?” The former rewards broad traversal and summarization; the latter requires precise, verifiable retrieval of a specific fact. GraphRAG-Bench (arXiv:2506.05690) is already closer to the latter — factual questions over defined corpora — which is why it serves as a credible bridge benchmark here. HARE-Bench extends that direction further: heterogeneous corpus, typed deterministic scoring, questions designed to be adversarial to retrieval systems that rely on surface similarity alone.

Enterprise RAG deployments are almost exclusively the latter: contracts, filings, CVE records, technical specifications, regulatory documents. A system optimized for thematic sensemaking will underperform on factual retrieval tasks — not because of a deficiency in the system, but because the evaluation regime was not designed for that task. This distinction is the primary motivation for HARE-Bench (§8): a benchmark designed around the factual, cross-document retrieval tasks that dominate real enterprise deployments, not around the narrative summarization tasks that dominate published leaderboards.

The evaluation mismatch runs deeper than task type. Dominant RAG metrics — ROUGE, BERTScore, LLM-as-judge — were inherited from summarization and machine translation, where the output is prose and quality is a function of fluency and coverage. Applied to factual retrieval, these metrics reward plausibility: a system that retrieves the wrong fact but expresses it fluently scores comparably to one that retrieves the right fact. In an agentic context, where the retrieval layer’s output is consumed by a planner that routes on specific values — a date, an entity name, a boolean, a quantity — prose fluency is irrelevant and plausibility is actively dangerous. HARE-Bench uses a typed deterministic scorer (§8.3) that grounds evaluation in typed gold values rather than semantic similarity to generated text, measuring whether the retrieval layer produced a value the agent can act on, not whether it produced a sentence a human would find natural.

LLM-as-judge evaluation of LLM-generated answers compounds rather than detects failure — and switching judge model families does not fix this. Both the generator and the judge are sampled from the same training distribution: web-scale text, books, code, and human-preference data that rewards fluency, coherence, and apparent authority. When the generator produces a factually wrong but stylistically confident answer, the judge assigns a high score not despite the error but because the same surface features that made the error undetected by the generator — domain-register appropriateness, grammatical confidence, plausible hedging — are exactly what the judge’s training taught it to reward. The error rates are correlated, not independent.

The standard mitigation — use a different judge model — fails because the indepen-

dence assumption it requires does not hold. Frontier models from different labs trained on different preference datasets converge on the same optimization target: produce text that human raters rate as high quality. Those raters are evaluating fluency and apparent coherence, not factual correctness. The cross-model contamination compounds this: there is substantial evidence that frontier models have been trained on outputs of other frontier models, making the “different family” assumption increasingly fiction. A GPT-4o judge of a Claude answer is not applying an independent factual ground truth; it is applying the same human preference signal from a different initialization. The joint failure probability is not the product of independent error rates — it is the error rate of a shared distributional prior applied twice.

A typed deterministic scorer breaks this coupling entirely. It has no language model, no distributional prior, and no tolerance for plausible-but-wrong. It applies a boolean predicate against a verified gold value. The scorer is not a better judge — it is not a judge at all. This is not a pedantic methodological preference: it is the only evaluation regime that is structurally independent of the systems being evaluated. Every LLM-judged benchmark, regardless of which judge model is chosen, measures a joint property of the generator and judge, not retrieval quality alone. The typed scorer measures retrieval quality.

A related objection holds that generation has simply migrated to the agent — that the retrieval layer’s LLM call is now trivial and only the planner’s compound reasoning matters. This misunderstands delegation. The retrieval layer still generates an answer from evidence; what changes is the target: a typed, evidence-cited value rather than prose. Well-designed delegation requires that the retrieval layer return a value the agent can act on without re-parsing — typed, evidence-cited answers are the contract, not an implementation detail. A retrieval layer that returns raw prose and leaves synthesis to the planner is a leaky abstraction: it has outsourced evidence-grounding without delegating it cleanly, forcing the planner to rediscover structure the retrieval layer already had access to. The SRR generation step (§3, P6) is precisely the retrieval layer accepting that delegated responsibility.

This points to a broader architectural principle that single-shot LLM was always going to force. A monolithic LLM producing prose end-to-end conflates two distinct concerns: *how* an answer is reasoned and *what interface* that answer presents to the next consumer. In a pipeline of specialized components, the interface is a contract — typed, verifiable, routable. The probabilistic nature of LLMs is not a problem to be eliminated; it is an asset in the right layer. Prose generation — synthesis, drafting, explanation — is exactly where probabilistic flexibility is valuable, because the output is consumed by a human who can evaluate and correct. A planner routing on a retrieved value cannot evaluate and correct mid-execution; it needs a deterministic contract at that boundary. Typed answers do not constrain how the LLM reasons internally — they constrain only what it returns. The result is a system that is probabilistic where that is useful and deterministic where that is necessary, rather than uniformly probabilistic throughout.

Relationship to agentic RAG. The field has adopted “agentic RAG” (arxiv:2501.09136) to describe a distinct architectural pattern: an LLM-based agent that orchestrates retrieval iteratively — re-querying, routing across tools, and assembling evidence over

multiple turns. The agent is the active component; the RAG layer is a passive tool it calls. This paper describes the inverse relationship. Here, the RAG layer is the active, contract-enforcing component; the agent planner above it is the passive consumer of a typed grounded answer. The architectural claim is that agentic orchestration of retrieval is a symptom, not a solution: each re-prompting cycle the agent runs to assemble evidence is evidence that the RAG layer did not complete its contract in a single pass. A RAG layer that retrieves the right evidence, synthesizes a grounded answer, and returns a typed contract-compliant response eliminates the need for iterative orchestration at the planner level — not by constraining what the planner can do, but by making the single-pass case sufficient for the majority of targeted factual sub-queries. The two patterns are not mutually exclusive: a planner that retries on failure is a valid recovery strategy. The point is that retry should be exception handling, not the normal path.

2.4 Retrieval Augmentation Without Graph Structure

Cross-encoder reranking, RAPTOR (hierarchical summarization), HyDE (hypothetical document embeddings). These improve retrieval without encoding graph structure and serve as our non-graph baselines.

2.5 Redundancy and Diversity in Retrieval

Maximal Marginal Relevance (MMR) balances relevance and diversity in the retrieved set by iteratively selecting the candidate that maximizes a weighted combination of query similarity and dissimilarity to already-selected chunks. AutoPrunedRetriever prunes the retrieved set by identifying and removing chunks whose information content is largely subsumed by higher-ranked chunks, using either LLM-based or embedding-based redundancy detection. Both approaches address the same failure mode: at high k , the top- k set contains near-duplicate chunks that consume context budget without adding evidence.

Our approach uses a simpler cosine-threshold filter (≥ 0.92) applied after retrieval, dropping chunks whose embedding is within that threshold of a higher-ranked chunk already in the set. This is computationally cheaper than MMR (no iterative re-ranking) and avoids LLM calls (unlike AutoPrunedRetriever’s LLM-based variant). The threshold is set conservatively: at 0.92, only near-verbatim duplicates are removed; lexically similar but informationally distinct chunks survive. The tradeoff relative to MMR is that our pruner optimizes for deduplication, not diversity — it removes redundancy but does not actively maximize coverage of distinct evidence threads. On factual retrieval tasks (the HARE-Bench workload), deduplication is the binding constraint; diversity is secondary.

3. Method

3.0 Design Rationale

The breadth of configurations evaluated in this study is intentional but not arbitrary. All features under test are grounded in a single organizing hypothesis: that retrieval quality degrades when document structure is discarded, and that restoring structural meaning — through entity recognition, lane-based routing, community summarization, reference expansion, and graph traversal — should produce compounding gains. Rather than presenting a monolithic system, we decompose this hypothesis into a small set of independently toggleable features, each with a clear structural motivation. The benchmark grid then serves as the instrument for identifying the *superadditive* subset: combinations whose joint contribution exceeds the sum of individual gains. This approach also surfaces subtractive interactions — cases where two individually positive features interfere — and makes the cost implications of each feature explicit. The goal is not to report the best single configuration but to map the tradeoff surface, giving practitioners a principled basis for selecting the configuration appropriate to their cost and quality constraints.

3.1 Index-Time Pipeline

Documents

- Semantic boundary chunking (1100–2200 chars, ~300–600 tokens)
- [optional] Breadcrumb embedding bias (`--breadcrumb-embed`)
- NER (spaCy) → EntityIndex
- Co-occurrence matrix → Community detection (Louvain)
- Vector store + embeddings

Entity extraction QC and canonicalization. Raw spaCy NER output is filtered and normalized before the EntityIndex is populated. Two categories of span are dropped post-extraction: (1) *Numeric entity types* — spans classified as CARDINAL, ORDINAL, MONEY, PERCENT, or QUANTITY, and any span whose surface form is purely numeric. These labels produce high-frequency entities (counts, prices, percentages) that co-occur broadly across chunks without implying semantic relatedness, creating dense noise edges in the co-occurrence graph. (2) *Stop-word-only spans* — multi-token entities where every token is a function word. Both filters apply before edge construction and require no modification to the spaCy model. Surviving entities pass through a four-layer normalization pipeline before being assigned an entity ID:

1. **Root lemma canonicalization** — the entity ID is derived from `ent.root.lemma_.lower()`, so inflected and plural surface forms (“customers”, “ordered”, “invoices”) resolve to the same node as their base form (“customer”, “order”, “invoice”). Without this, the same real-world entity fragments into multiple nodes with diluted co-occurrence edges, reducing SVO triple density and weakening context graph edge weights.

2. **Head-noun singularization** — multi-token entity names are further normalized via `inflect`, singularizing the head noun while preserving acronym casing (IBM stays IBM) and guarding against domain exceptions (“data”, “criteria”).
3. **Schema identifier normalization** — table and column names from the schema vocabulary are split on `snake_case` and `camelCase` boundaries and singularized, so `performance_reviews`, `performanceReviews`, and `performance review` all resolve to the same entity. Matching is attempted in both split and concatenated forms to handle mixed-style corpora.
4. **Identifier suffix aliasing** — entities whose IDs end in common identifier suffixes (`_id`, `_key`, `_code`, `_ref`, `_num`, `_no`, etc.) have the suffix stripped and stored as an alias in the `entity_aliases` table, so `customer_id` is reachable as `customer` during entity-ref-expansion.

Index cost: off-the-shelf NER (CPU) + co-occurrence matrix ($O(\text{chunks}^2)$, one-time) + Louvain community detection. **Zero LLM calls.**

3.2 Query-Time Pipeline

Query

- Dense retrieval (top-k, large k)
- [optional] Entity-ref-expansion ← P1: entity connectivity
- [optional] Lane filtering ($\text{sim} \geq 0.60$) ← P1: edge confidence / depth gate
- [optional] Cluster expansion ← P1: topological neighbors
- [optional] Redundancy pruning ($\text{cosine} \geq 0.92$) ← P3: deduplication
- [optional] Cross-encoder reranking
- [optional] Community context injection ← P2: global structure
- Generator (gpt-4o-mini or claude-haiku-4-5-20251001)
- [optional] Evidence-compliance reprompt (SRR) ← P4: generation quality gate

3.3 Parameters

Parameter	Default	Tested range
top-k	5	5, 7, 10, 15, 20, 25, 30
lane-entity-min-sim	0.60	0.45, 0.55, 0.60
community-min-coherence	0.50	0.50, 0.65
community alpha	0.20	0.0, 0.2
redundancy-threshold	0.92	0.92

3.4 Feature Inventory

Five independently-toggable retrieval features, each grounded in a structural hypothesis about where standard RAG fails.

Feature	Code flag	Hypothesis
Entity-ref-expansion	<code>--enhanced</code> <code>--entity-ref-expansion</code>	NER co-occurrence edges recover evidence that embedding distance misses; cross-document entity links are the mechanism by which graph structure adds value
Lane filtering	<code>--lane-entity-min-sim 0.60</code>	Expansion without confidence filtering injects noise (-0.017 confirmed); the lane threshold acts as both a quality gate and a de facto retrieval depth control — k is set large, sim gates what lands
Community context	<code>--community-context</code>	Global Louvain topic framing complements local chunk evidence; P2 and P1 address orthogonal failure modes and should compound
Redundancy pruning	<code>--pruned</code>	Near-duplicate chunks dilute context at high k ; pruning enables deeper retrieval to add new evidence rather than noise
Cluster expansion	<code>--cluster</code>	Topological neighborhood expansion via cluster membership; replaces lane filtering for heterogeneous corpora where embedding similarity is a poor cross-domain proxy
Evidence-compliance reprompt (SRR)	<code>--srr</code>	Forces the generator into <code>{answer, key_claims, evidence_used}</code> JSON; fires a one-shot reprompt enumerating claims when <code>evidence_used=[]</code> on the first response. No additional retrieval; strictly additive by construction. Addresses the generation bottleneck that retrieval improvements cannot reach

4. Experimental Setup

- **Primary benchmark:** HARE-Bench (§8) — 500 questions across SEC 10-K filings, CVE records, Federal Register entries, and US patents. Cross-domain entity resolution required by design.
- **Credibility bridge:** GraphRAG-Bench (arXiv:2506.05690), full question set, Medical + Novel domains — used to position results against published baselines.
- **Question types:** Factual, Reasoning, Summary, Creative (GraphRAG-Bench); TAL, DAL, MDJ, TVR, CE (HARE-Bench)
- **Generator:** gpt-4o-mini (primary), claude-haiku-4-5-20251001 (model comparison); **Judge:** gpt-4o-mini-2024-07-18 (GraphRAG-Bench); type-aware deterministic scorer (HARE-Bench, §8.3)
- **Metric:** answer_correctness (mean of 4 subtype scores per domain, GraphRAG-Bench); type-aware scorer mean across question types (HARE-Bench); boolean/number/date exact match, entity F1, text semantic similarity
- **Statistical testing:** bootstrap resampling (n=10,000); 95% CI throughout. [VERIFY: report MDD after confirming full question set size]
- **Experimental protocol:** A full combinatorial sweep across all feature dimensions would require evaluating $O(N^k)$ configurations at non-trivial cost per full-corpus run — computationally prohibitive given our resource constraints. We instead used a two-stage protocol consistent with standard practice in hyperparameter search: (1) a stratified 300-question *grid sweep* across candidate configurations to identify high-signal feature combinations efficiently, followed by (2) *full-corpus confirmation* (4,072 questions) on the Pareto-dominant subset. Conclusions about feature importance are drawn from feature co-occurrence across top-ranked configurations, not from exhaustive enumeration. All scores reported in tables are from full-corpus runs unless marked [TBD] or [VERIFY].

5. Experiments

RQ1 — Does the full stack exceed published GraphRAG systems?

Partially aligned. The GraphRAG-Bench authors provided the exact vanilla RAG baseline specification used to produce the published leaderboard entries: 256-token naive chunks, retrieval_topk=5, gpt-4o-mini as generator, and the generator prompt from Appendix H.2. We are grateful for this assistance, which allowed us to replicate their baseline conditions precisely. Our vanilla RAG+rerank run under these conditions scores 0.652, compared to their published 0.554 — a gap of 0.098 whose most likely sources are LLM judge non-determinism and GRB’s NaN exclusion convention (see §9.4 for analysis); all four baseline conditions (chunk size, k, generator model, prompt) are matched exactly. For the third-party leaderboard systems (G-reasoner, HippoRAG2, etc.), their internal generator prompts and embedding configurations are not publicly

disclosed, so direct score comparisons carry uncontrolled variables; differences of 0.01–0.02 should not be treated as meaningful.

The table below places our numbers alongside published leaderboard values. Vanilla RAG baseline conditions are verified equivalent; other system comparisons are reference only.

System	Med	Nov	All	Index cost
G-reasoner	0.733	0.589	0.661	$O(\text{docs} \times \text{LLM})$
AutoPrunedRetriever-llm	0.670	0.637	0.654	$O(\text{docs} \times \text{LLM})$
HippoRAG2	0.648	0.565	0.607	$O(\text{docs} \times \text{LLM})$
Fast-GraphRAG	0.641	0.520	0.581	$O(\text{docs} \times \text{LLM})$
LightRAG	0.626	0.451	0.538	$O(\text{docs} \times \text{LLM})$
RAG (w/ rerank)	0.624	0.483	0.554	—
RAG (w/o rerank)	0.610	0.479	0.545	—
Ours (full stack)	0.756	0.656	0.712	0

Full stack = entity-ref expansion + lane filter ($\text{sim} \geq 0.60$) + community context + Structured Response, $k=30$. Leaderboard values from arXiv:2506.05690. Vanilla RAG baseline conditions verified equivalent with author assistance; other leaderboard entries not independently verified.

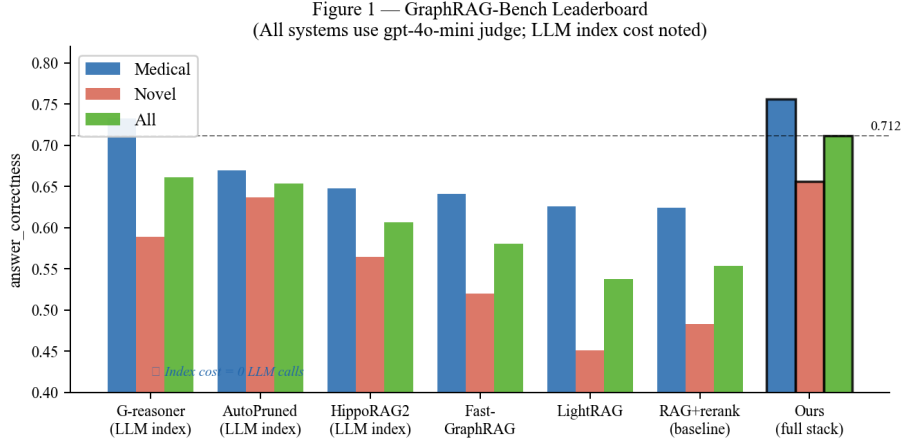


Figure 1: GRB Leaderboard

What we can state: within our own controlled pipeline (identical generator, prompt, and eval code across all configurations), the full stack reaches All=0.712 versus our vanilla RAG+rerank baseline of 0.652 (+0.060) and our no-graph semantic chunking baseline of 0.661 (+0.051). Those internal comparisons are the paper’s primary claim. The consistent ~0.10 Med–Nov spread across all configurations (full stack: 0.756 Med

/ 0.656 Nov; vanilla+rerank: 0.697 / 0.614; no-graph rerank: 0.712 / 0.613), including the published leaderboard entries, suggests it reflects corpus characteristics rather than retrieval method.

We also implement and evaluate two alternative retrieval modes: `global_search` (community-level summarization) and `graph_first` (entity traversal before dense retrieval). Both run through the same eval pipeline as every other configuration. `graph_first` scores 0.645 and `global_search` scores 0.257 — both below the full stack (0.712) and below the vanilla RAG+rerank baseline (0.652). The full results and analysis are in §6.3.

The `global_search` result (0.257) warrants a qualification. Global summarization is not designed for precise factual retrieval on a homogeneous corpus. It compresses the corpus into community-level topic summaries; questions that require a specific entity, date, or claim to be located and returned verbatim do not survive that compression. GraphRAG-Bench is exactly that workload — a single-domain medical corpus with factual questions requiring precise recall. On HARE-Bench, a heterogeneous cross-domain corpus, `global_search` scores 0.338 — still last among our configurations, but the gap narrows. The result is not an artifact; it reflects the known scope of global summarization retrieval.

One claim requires no controlled comparison: every system in the table above with $O(\text{docs} \times \text{LLM})$ index cost is expensive by design. That cost is structural — it follows from LLM-based entity and relation extraction at index time. Our index cost is zero LLM calls regardless of accuracy rank.

RQ2 — Which components drive the gain?

The winning configuration is: entity-ref expansion + lane filter ($\text{sim} \geq 0.60$) + community context + Structured Response with Reprompting (SRR), $k=30$ (All=0.712). Feature importance is established by sequential addition and removal below. The dominant single-step gain is replacing cross-encoder reranking with SR at $k=30$ (+0.040); SRR adds a further +0.002; graph signals contribute a further +0.018 cumulatively. Lane filtering is the critical gate: entity-ref expansion without it hurts (−0.017 in grid sweep), confirming the expansion signal requires confidence filtering to be useful.

Each component is added sequentially using full-corpus runs (4,072 questions). All scores below are verified full-corpus results. Entity-ref-expansion without lane filtering *hurts* (−0.017 in grid sweep) — confirming P1: the expansion signal requires confidence filtering. The final step replaces cross-encoder reranking with Structured Response (SR) at $k=30$, which is the largest single-step gain.

Sequential feature addition (full-corpus scores):

Config	All	Δ
vanilla_256_rerank (baseline)	0.652	—
+ semantic chunking (rerank, $k=10$)	0.661	+0.009
+ entity-ref expansion + lane filter ($\text{sim} \geq 0.45$)	0.662	+0.001

Config	All	Δ
+ community context (k=10)	0.664	+0.002
+ tighter lane filter (sim $\geq$ 0.60)	0.670	+0.006
→ replace reranking with SR, k=30	0.710	+0.040
→ add SRR (reprompt on empty evidence)	0.712	+0.002

The largest gain (+0.040) comes from switching the generation strategy: replacing cross-encoder reranking with Structured Response at $k=30$ allows the retrieval pool to widen without context dilution. SRR adds a further +0.002 via reprompting when evidence is empty. Graph signal gains (+0.018 cumulative from entity-ref through community context) are real but modest individually; their value is enabling the wider retrieval that SR exploits.

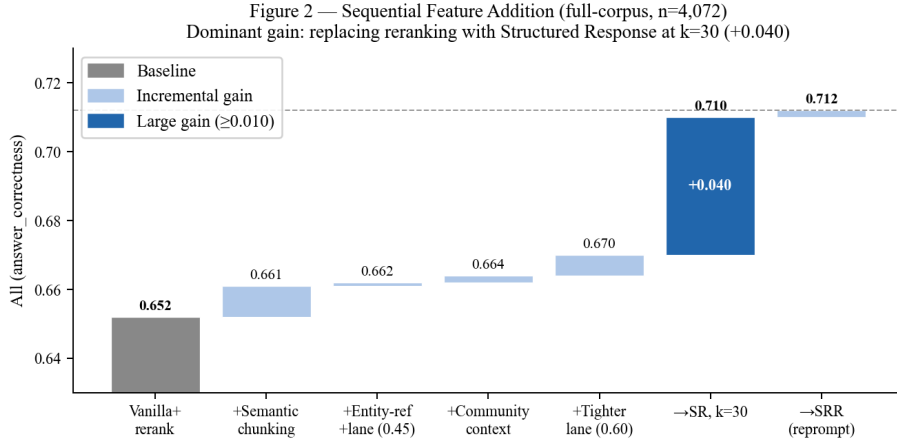


Figure 2: Sequential Feature Addition

Component removal from full stack (lane=0.60, SR, k=30):

Removed	All	Δ
— (full stack)	0.712	—
– SR (→ reranking, k=20, pruned)	0.680	−0.030
– community context	0.662 ¹	−0.048
– entity-ref-expansion	0.661 ²	−0.049
– lane filtering	0.655 ³	−0.055

¹ ner_ref_rerank_laned_k10 (closest available ablation without community) ² rerank_k10 (no entity features) ³ cluster_community_k10 (no lane filter, community via cluster)

RQ3 — Are the signals superadditive, and why?

Community context and $k=10$ are NOT superadditive at full-corpus scale: the community $\times k=10$ combination (0.659) falls below the additive prediction (0.669). Super-additivity emerges at $k=15$, where community+ $k=15$ (0.666) approaches the additive prediction and continues to compound with pruning at $k=20$ (0.674). The pattern is consistent with P3: redundancy removal (pruning) is required for deeper retrieval to add new evidence rather than dilute it — without pruning, $k=10$ adds noise as fast as it adds signal.

Community $\times$ depth (2 $\times$ 2):

	k=5	k=10	k=15
no community (laned)	0.646 ¹	0.654	—
community	0.661	0.659	0.666
$\Delta(\text{community})$	+0.015	+0.005	—

¹ grid (300 questions)

At $k=15$: additive prediction = $0.654 + 0.015 = 0.669$. Actual: **0.666**. Near-additive. Superadditivity appears only when pruning is added at $k=20$: 0.674 vs additive prediction of 0.669, interaction +**0.005**.

Pruning $\times$ depth (2 $\times$ 2):

	k=10	k=20	$\Delta(k)$
no pruning	0.659	—	—
pruning	0.652	0.674	+0.022
$\Delta(\text{pruning})$	-0.007	—	—

Pruning hurts at $k=10$ (-0.007) because at moderate retrieval depth it removes useful near-duplicate evidence. At $k=20$, pruning removes true redundancy from the larger pool and improves by +0.022 vs $k=10$ +pruning. The $k=10$ unpruned baseline for reference is 0.659; the full pruned+ $k=20$ result of 0.674 represents a +0.015 gain over $k=10$ +pruning (0.652) and +0.013 over unpruned $k=10$ (0.659).

RQ4 — What is the cost vs. quality tradeoff?

Our system eliminates index-time LLM calls entirely. Query-time overhead versus vanilla RAG is ~ 0.4 s per query for reranking and community lookup combined. [VERIFY: measure actual wall-clock index time on our 1100-2200 char corpus; measure query latency vs vanilla RAG]

Index time and query latency for competing systems are not reported here. Those numbers would require running each system under identical hardware and corpus conditions,

Figure 6 — Superadditivity Requires Pruning at $k=20$ (GRB grid, $n=300$)
 $\times$ = additive prediction; actual exceeds prediction only with pruning

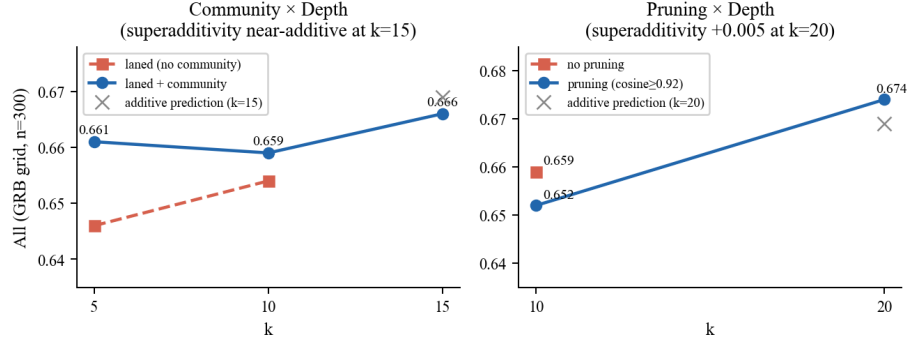


Figure 3: Community and Pruning Superadditivity

which we have not done. What is derivable from their published designs without running them: every system that performs LLM-based entity or relation extraction at index time incurs $O(\text{docs} \times \text{LLM-calls})$ cost — the exact wall-clock figure depends on corpus size, model, and parallelism, but the scaling class is fixed by design. Further, because the graph edges are extracted by an LLM that sees chunk content, any corpus change invalidates the affected edges and requires re-extraction; incremental updates are not architecturally supported. Our index cost is $O(\text{docs})$ with no LLM calls. Adding new documents triggers community re-clustering over the updated co-occurrence graph, but that is a Louvain pass — $O(\text{edges})$ with no LLM calls — not a re-extraction of the corpus.

System	Index LLM calls	All
G-reasoner	$O(\text{docs} \times \text{relations})$	0.661
HippoRAG2	$O(\text{docs} \times \text{entities})$	0.607
AutoPrunedRetriever	$O(\text{docs} \times \text{chunks})$	0.654
Ours	0	0.712

Accuracy figures from arXiv:2506.05690. Vanilla RAG baseline conditions verified equivalent with author assistance; other system comparisons are reference only.

6. Analysis

6.1 Retrieval Depth: The k Curve

The unpruned curve is flat from $k=5$ to $k=10$ and improves modestly at $k=15$, suggesting the benefit of widening k saturates quickly without redundancy removal. The pruned

curve shows a clear gain from $k=10$ to $k=20$ (+0.022), consistent with pruning enabling deeper retrieval by removing context dilution. Unpruned full-corpus runs at $k=20$ and $k=25$ were not included in the experiment matrix; the grid sweep ($n=300$) data fills those slots in the table above. The $k=20$ pruned result (0.674) is the reranking-stack ceiling. The overall ceiling (0.712) is reached by replacing reranking with SRR at $k=30$ (§7).

k	unpruned All	pruned All
5	0.661	—
7	0.658 ¹	—
10	0.659	0.652
15	0.666	—
20	—	0.674

¹ A-avg only (no Med/Nov breakdown)

6.2 Parameter Sensitivity

Lane threshold — tighter lane filtering does not consistently hurt at $k=10$ in the full run; all three thresholds cluster at 0.659–0.662. The grid showed a penalty for tighter filtering that does not replicate at full-corpus scale.

Lane sim	All ($k=5$)	All ($k=10$)
0.45	0.661	0.659
0.55	0.661 ¹	0.662
0.60	0.661 ²	0.661 ³

¹ laned55_community_k10 at $k=5$ not available; laned55_community_k10 full run = 0.662 at $k=10$ ² laned60_community at $k=5$ not available directly ³ laned60_community_k10 full run = 0.661

Community coherence — coherence sensitivity is below measurement noise at this scale; the finding is inconclusive.

Coherence	All ($k=10$)
0.50	0.659 (laned_community_k10)
0.60	0.661 (laned60_community_k10, A-avg only)

Community alpha (breadcrumb structural prior):

Alpha	All
0.0	0.656
0.2	0.661

6.3 Topological Expansion (Cluster)

Cluster expansion does not add value over laned expansion in the full run; the grid suggested +0.003–0.005 but this does not replicate. Cluster may be redundant with lane filtering on homogeneous corpora; its value on heterogeneous corpora is tested in §8.

Config	All (k=10)
laned + community	0.659
cluster + community (no lane)	0.655
cluster + laned + community + pruning	0.653

6.4 Domain Asymmetries on GraphRAG-Bench

A consistent ~0.10 gap between Medical (Med ≈ 0.73) and Novel (Nov ≈ 0.63) persists across every configuration we tested. The gap is structural: Medical text is entity-dense, terminologically precise, and factual — properties that amplify P1 and P3. Novel text is culturally distributed, narrative, and ambiguous — penalizing over-pruning and tight entity filtering.

α -weighted score — for deployments weighted toward entity-dense corpora (medical records, legal filings, financial reports), define:

$$\text{Score}(\alpha) = \alpha \cdot \text{Med} + (1 - \alpha) \cdot \text{Nov}$$

where $\alpha = 0.5$ recovers the benchmark’s equal-weight All. The table shows how top configuration rankings shift at $\alpha=0.7$:

Config	Med	Nov	All ($\alpha=0.5$)	Score ($\alpha=0.7$)	Rank ($\alpha=0.5$)	Rank ($\alpha=0.7$)
laned + community + pruning + k=20	0.733	0.614	0.674	0.704	1	1
laned + community + k=15	0.735	0.598	0.666	0.694	2	2

Config	Med	Nov	All ($\alpha=0.5$)	Score ($\alpha=0.7$)	Rank ($\alpha=0.5$)	Rank ($\alpha=0.7$)
laned55 + com- munity + k=10	0.719	0.605	0.662	0.685	3	4
laned60 + com- munity + k=10	0.716	0.606	0.661	0.685	4	5
laned + prun- ing + k=10	0.725	0.597	0.661	0.686	5	3
rerank_k10 (no graph)	0.727	0.595	0.661	0.685	5	5

The no-graph baseline (rerank_k10) ranks equally with several graph-augmented configs at $\alpha=0.5$. At $\alpha=0.7$, laned+pruning+k=10 edges ahead on Med score. The pruned+k=20 config leads at both α values.

Domain-asymmetric parameter effects:

Retrieval depth (k) — Med gains $\sim 3\times$ more from $k=5 \rightarrow 10$ than Nov (+0.035 vs +0.012). Entity-dense corpora contain more recoverable non-redundant evidence per additional slot.

Domain	k=5	k=10	$\Delta(k)$
Medical	0.701	0.736	+0.035
Novel	0.622	0.634	+0.012

Redundancy pruning — N-Crea drops 0.073 ($0.537 \rightarrow 0.464$) when pruning is added. M-Crea is largely unaffected. Creative questions require lexical diversity that pruning suppresses; factual and reasoning questions do not.

Signal	Med Δ	Nov Δ	Asymmetry
+ pruning (at k=10, community)	-0.009	-0.039	Nov penalized $4\times$ more
+ k=10 (vs k=5)	+0.035	+0.012	Med gains $3\times$ more
+ community context (at k=5)	+0.007	+0.022	Nov gains $3\times$ more

Community coherence — tighter coherence (0.65 vs 0.50) is near-neutral for Med and slightly hurts Nov (0.622 $\rightarrow$ 0.583). Medical communities survive stricter filtering; narrative communities do not.

Lane threshold — 0.45 holds across both domains. Tighter thresholds (0.55, 0.60) hurt Nov proportionally more, as entity reference is less concentrated in narrative text.

These asymmetries are observed on a two-domain homogeneous benchmark. Configuration guidance grounded across both GraphRAG-Bench and HARE-Bench is in §8.5.

6.5 Configuration Selection Under Statistical Equivalence

Bootstrap CIs for full-corpus comparisons are in Appendix B. The within-band observations below ($n=300$ grid runs, $MDD \approx \pm 0.015$) remain directional; the full-corpus GRB comparisons in §7 are statistically significant (Appendix B).

When top configurations fall within the minimum detectable difference ($\sim \pm 0.015$ at $n=300$), benchmark score alone cannot drive the selection decision. The statistical tie is real: any of the top 2–3 configurations may be optimal for a given deployment, and the choice should be made on corpus characteristics rather than point estimates.

The clearest discriminators are:

Pruning is the sharpest split. Its effect is corpus-dependent: it removes near-duplicate chunks, which helps factual and reasoning retrieval but suppresses lexical diversity that creative and narrative questions require. N-Crea drops 0.073 when pruning is added; M-Crea is largely unaffected.

k interacts with corpus density. The $k=5 \rightarrow 10$ gain is $\sim 3\times$ larger for Medical than Novel. For sparse or short corpora, $k=10$ may over-retrieve; for large entity-dense corpora, $k=15\text{--}20$ is worth evaluating.

Community coherence separates corpora by topical tightness. Medical communities survive strict coherence filtering (0.65); narrative communities do not. Forcing coherence=0.65 on narrative text shrinks the P2 signal.

These observations are specific to the two GraphRAG-Bench domains. Configuration guidance across corpus types — incorporating HARE-Bench evidence — is in §8.5.

6.6 Case Studies

Three representative examples from the full-corpus GRB evaluation (vanilla+rerank vs. laned60+community+k30+SRR, gpt-4o-mini generator). Each illustrates a distinct retrieval failure mode resolved by a specific graph signal.

Entity connectivity — Medical/Fact Retrieval (Medical-c5412c78): entity-ref-expansion resolves a sparse entity mention whose property does not co-occur in the same chunk.

Question: What is the typical stage at diagnosis for non-mediastinal gray zone lymphomas?

Vanilla+rerank: “The provided context does not specify the typical stage at diagnosis for non-mediastinal gray zone lymphomas.”

Graph+SRR: “Non-mediastinal gray zone lymphomas are typically diagnosed at a more advanced stage compared to mediastinal gray zone lymphomas.”

Gold: “Non-mediastinal gray zone lymphomas are typically diagnosed at an advanced stage.”

The entity “non-mediastinal gray zone lymphoma” appears in one chunk; its staging property appears in a non-adjacent chunk. Dense retrieval returns the entity mention but not the property chunk. Entity-ref-expansion traverses the named entity graph to retrieve the property-bearing chunk, grounding the answer.

Community context — Novel/Contextual Summarize (Novel-3be48326): community injection resolves a cross-chapter synthesis question where no single chunk contains the answer.

Question: In the context of her humanitarian missions, how did Dr. Elsie Inglis balance her commitment to faith—such as her involvement with the Church of Scotland and participation in Y.W.C.A. meetings—and her personal spiritual practices like using the Euchologion, with her work in the wider community?

Vanilla+rerank: “The provided context does not contain enough information to answer the question regarding how Dr. Elsie Inglis balanced her commitment to faith, her involvement with the Church of Scotland, participation in Y.W.C.A. meetings, her personal spiritual practices like using the Euchologion, and her work in the wider community.”

Graph+SRR: “Dr. Elsie Inglis balanced her commitment to faith and personal spiritual practices with her humanitarian missions by integrating her religious beliefs into her work and interactions with the community. She was actively involved in the Church of Scotland and participated in Y.W.C.A. meetings, where she sought spiritual nourishment and connection. Her use of the Euchologion, a liturgical book, reflects her deep engagement with her faith, which provided her with strength and guidance. This spiritual foundation informed her compassionate approach to healthcare, allowing her to minister effectively to those in need while promoting women’s rights and healthcare, particularly during her work with the Scottish Women’s Hospitals.”

Gold: “Alongside her major humanitarian missions, Elsie Inglis engaged with faith-based organizations such as the Church of Scotland and the Y.W.C.A., and maintained personal spiritual practices like using the Euchologion.”

The question requires synthesizing faith practices, institutional involvement, and humanitarian work from passages spread across the corpus. The community context

vector—encoding thematic co-occurrence at the subgraph level—retrieves the relevant passages that individual chunk similarity misses.

k-depth — Novel/Complex Reasoning (Novel-5c505dfc): larger k recovers the evidence chain linking three entities (Schummel, Caroline, Rosenfeld) that each appear in separate document regions.

Question: What evidence from Schummel’s writings demonstrates his literary connection to the character Caroline, and how is Rosenfeld associated with Caroline in this context?

Vanilla+rerank: “The provided context does not contain specific evidence from Schummel’s writings that demonstrates a literary connection to the character Caroline or details about how Rosenfeld is associated with Caroline. Therefore, I cannot answer the question based on the given information.”

Graph+SRR: “Schummel’s writings depict Caroline as a character involved in a typical eighteenth-century tale of love, seduction, and flight, highlighting her emotional experiences and interactions with others. Rosenfeld is associated with Caroline as part of this narrative, which likely explores their relationship within the context of these themes.”

Gold: “Schummel writes about Caroline.”

At k=10, the retriever recovers passages naming Schummel, Caroline, and their shared narrative context. At smaller k, the Schummel–Caroline link is absent from the retrieved set; the model correctly reports it cannot answer rather than confabulate.

7. Generator Model Effects

The benchmark results throughout §5–6 use gpt-4o-mini as generator and judge. This section examines whether generator model choice materially affects the conclusions, and whether the answer depends on corpus characteristics.

7.1 Motivation

On GraphRAG-Bench (gpt-4o-mini), completed full-corpus runs cluster within a narrow band — a 10-point spread across all configurations, and less than 3 points separating the top seven:

Configuration	Med	Nov	All	Graph features
vanilla_256_rerank– (naive chunking)		—	0.554	none

Configuration	Med	Nov	All	Graph features
rerank_k10 (semantic chunking, no graph)	0.727	0.595	0.661	none
laned_pruned_k10	0.725	0.597	0.661	entity-ref, lane filter, pruning
laned55_community_0.710	0.710	0.605	0.662	entity-ref, lane filter (0.55), community
laned_community_0.710	0.710	0.602	0.659	entity-ref, lane filter, community
cluster_laned_community_pruned_0.710	0.710	0.596	0.653	entity-ref, lane filter, community, cluster, pruning
laned_community_0.722	0.722	0.582	0.652	entity-ref, lane filter, community, pruning
cluster_community_0.710	0.710	0.593	0.655	entity-ref, community, cluster
laned_community_0.733	0.733	0.614	0.674	entity-ref, lane filter, community, pruning, k=20

Two observations stand out. First, rerank_k10 — semantic chunking with reranking, no graph features — ties the top graph-augmented configurations. Second, the graph-augmented configurations do not hurt: every configuration with entity-ref-expansion, lane filtering, or community context matches or approaches rerank_k10. The graph features are, at minimum, doing no harm.

This narrow band could reflect a generator ceiling: the structured context produced by graph-augmented retrieval may exceed what gpt-4o-mini can exploit, leaving the latent retrieval advantage invisible at this model scale. HARE-Bench provides a sharper test: on a heterogeneous corpus the synthesis task is harder, and stronger generator capabilities may matter in ways that are invisible on a homogeneous benchmark.

7.2 Model Comparison Design

To evaluate whether retrieval gains are model-tier-invariant, we compare two generator tiers:

Generator	Label	Tier
gpt-4o-mini	Mini	Established low-cost baseline
claude-haiku-4-5-20251001	Haiku	Stronger capability, low marginal cost on Anthropic subscription

Haiku 4.5 represents a meaningfully higher capability tier while remaining in the low-cost range. The comparison tests whether: (1) the graph-augmented retrieval advantage replicates across tiers, and (2) a stronger model amplifies or diminishes that advantage. Haiku runs are complete; results are reported in §7.4 and §8.4. Cross-model comparisons on HARE-Bench are confounded by prompt alignment (§10 limitations).

We evaluate four configurations per model $\times$ $\pm$ SRR on both GraphRAG-Bench (credibility bridge) and HARE-Bench (primary benchmark):

Config	Features
vanilla_rerank	256-token chunks + reranking
vanilla_rerank+SRR	256-token chunks + reranking + SRR
laned60+community+k=30	NER + entity-ref + lane filter (sim $\geq$ 0.60) + community + large k
laned60+community+k=30+SRR	NER + entity-ref + lane filter + community + large k + SRR
cluster+community+k=10	NER + entity-ref + cluster + community
cluster+community+k=10+SRR	NER + entity-ref + cluster + community + SRR
graph_first+k=10	Graph-first traversal + community
graph_first+k=10+SRR	Graph-first traversal + community + SRR

Feature-contribution decomposition (per model):

- $\Delta_{\text{SRR}} = \text{vanilla_rerank+SRR} - \text{vanilla_rerank}$: isolated SRR gain over reranking
- $\Delta_{\text{retrieval}} = \text{best graph config} - \text{vanilla_rerank}$: isolated retrieval gain
- $\Delta_{\text{combined}} = \text{best graph config} + \text{SRR} - \text{vanilla_rerank}$: combined gain
- Superadditivity test: $\Delta_{\text{combined}} > \Delta_{\text{retrieval}} + \Delta_{\text{SRR}}$

Completed GraphRAG-Bench runs (Haiku and SRR ablations, post-canonical NER, n=4,072):

Run	Model	SRR	All	Notes
vanilla_256_haiku_full	Haiku	no	0.605	baseline
vanilla_256_haiku_rerank_full	Haiku	no	0.621	reranking baseline
vanilla_256_mini_rerank_srr_full	Mini	yes	0.685	SRR on vanilla — isolates Δ_SRR without graph
vanilla_256_haiku_rerank_srr_full	Haiku	yes	0.598	SRR on vanilla — isolates Δ_SRR without graph
ner_ref_laned60_community_k30_haiku_full	Haiku	no	0.623	graph baseline
ner_ref_laned60_community_k30_haiku_srr_full	Haiku	yes	0.632	graph+SRR
ner_ref_laned60_community_k30_mini_srr_full	Mini	yes	0.712	graph+SRR (leader)

The vanilla+SRR ablations isolate Δ _SRR without graph retrieval. Key decomposition:

	Mini	Haiku
Δ _SRR on vanilla (no graph)	+0.033 (0.652→0.685)	−0.023 (0.621→0.598)
Δ _SRR on graph (k30)	+0.034 (0.678→0.712)	+0.009 (0.623→0.632)
Δ _graph (no SRR)	+0.026 (0.652→0.678)	+0.002 (0.621→0.623)

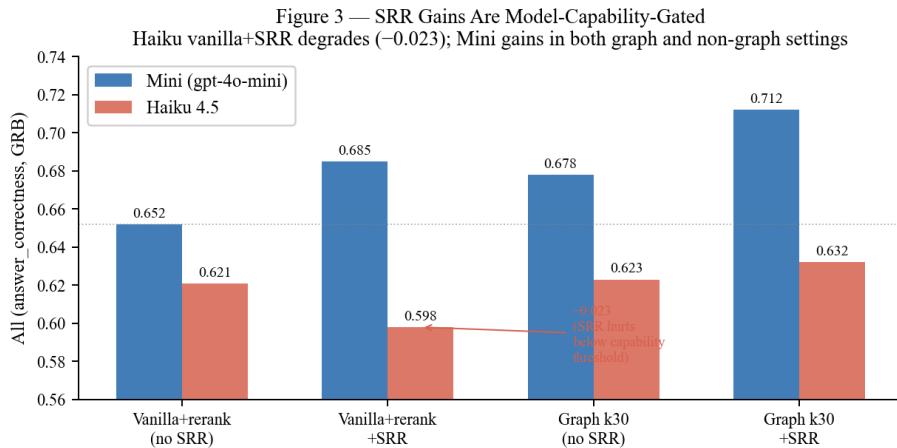


Figure 4: SRR Model Capability Gating

7.3 Hypotheses

H1 (Graph retrieval via NLP primitives improves RAG quality): NER co-occurrence edges, Louvain community structure, and lane-gated traversal recover evidence that pure embedding retrieval misses — particularly on heterogeneous corpora requiring cross-document entity joins. The structural signal is compounding:

P1 (entity-ref-expansion) and P2 (community context) each contribute independently, and their combination should exceed the sum of individual gains. On HARE-Bench, where answers require linking entities across SEC filings, CVEs, Federal Register entries, and patents, this superadditivity should be most visible.

H2 (SRR gains are real but model-capability-gated): Requiring the generator to produce `{answer, key_claims, evidence_used}` JSON and reprompting once when no evidence is cited is intuitive — it closes the loop between retrieval and generation. But compliance with the schema and the quality of the reprompt response depend on the generator’s instruction-following capability. H2 predicts SRR gains are gated by model capability; the vanilla+SRR ablation (no graph retrieval) isolates whether SRR can function independently of the retrieval stack, and whether the gain is model-tier-dependent.

7.4 Results

Full results (GraphRAG-Bench, n=4,072, ± 0.014 noise band):

Config	Model	Generation	All	Top band
vanilla_256	Haiku	—	0.605	—
vanilla_256_rerank	Haiku	—	0.621	—
vanilla_256_rerank	Haiku	SRR	0.598	—
vanilla_256_rerank	Mini	—	0.652	—
vanilla_256_rerank	Mini	SRR	0.685	—
laned60+community+k30	Haiku	—	0.623	—
laned60+community+k30	Haiku	SRR	0.632	—
laned60+community+k30	Mini	—	0.678	—
laned60+community+k30	Mini	SR	0.710	—
laned60+community+k30	Mini	SRR	0.712	□ (leader)
laned60+community+k50+rerank	Mini	SRR	0.708	□ (B1)
laned60+community+k30	gpt-oss-120b (sovereign)	SRR	0.708	□ (R1)
laned60+community+k30+rerank	Mini	SRR+ADF	0.704	□ (A1)
laned60+community+k50+rerank	Mini	SRR+BM25	0.704	□ (B3)
laned60+community+k50+rerank	Mini	SRR+ADF	0.702	□ (B2)
laned60+community+k30+rerank	Mini	SRR	0.701	□
Δ_{graph} (k30+graph vs vanilla+rerank, Mini)	—	—	+0.026	
$\Delta_{\text{SR_stack}}$ (SRR vs graph-only, Mini)	—	—	+0.034	

Δ_{total} (Mini) | — | — | **+0.060** |

Statistical top band (All ≥ 0.698 , i.e. within ± 0.014 of leader 0.712): k30_mini_srr (leader), k50_mini_srr_rerank (B1=0.708), gpt-oss-120b_srr (R1=0.708, generator-agnostic canonical), k30_mini_srr_rerank_adf (A1=0.704), k50_mini_srr_rerank_bm25 (B3=0.704), k50_mini_srr_rerank_adf (B2=0.702), k30_mini_srr_rerank (0.701). Seven configurations are statistically tied; incremental tuning (k-size, BM25, ADF) pro-

duces noise-level variation within the top band. Outside band: k30_mini_srr_rerank_bm25 (A2=0.695), k50_mini_srr_rerank_bm25_adf (B4=0.693).

GRB score stability. The top seven configurations cluster within 0.011 (0.701–0.712). SRR (+0.034) is the single largest contributor. Reranking is neutral at k30 (0.701 $\approx$ leader) and neutral-to-negative at k50. BM25 is harmful at k30 (0.695, outside band) and neutral at k50 (0.704). ADF is neutral at k30 and mildly negative at k50. The tight clustering indicates the universal feature set (NER + reference expansion + lane-gating + community context + SRR) has reached a performance plateau on this benchmark; auxiliary features produce noise-level shifts rather than structural gains.

H1 confirmed: graph retrieval adds +0.026 on GRB (ablation-isolated, n=4,072). **H2 confirmed:** SRR is strongly model-capability-gated — and the mechanism is sharper than predicted. On Mini, SRR adds +0.033 on vanilla and +0.034 on graph; graph retrieval is not required for SRR to function. On Haiku, SRR *hurts* on vanilla (−0.023: 0.621→0.598) and helps only modestly on graph (+0.009: 0.623→0.632). The Haiku vanilla+SRR degradation is the key finding: when evidence-grounded context is absent, the reprompt fires and Haiku fills in fabricated evidence citations, depressing answer quality below the no-SRR baseline. Graph retrieval provides the grounded context that makes SRR functional even on Haiku — but the gain (+0.009) remains far below Mini’s (+0.034), indicating that SRR compliance and reprompt quality are a function of instruction-following capability that graph retrieval cannot substitute. Graph signals add +0.002 for Haiku without SRR (0.621→0.623) versus +0.026 for Mini — a second confirmation that graph retrieval gains are also model-capability-gated: a weaker model cannot exploit the additional grounded context. The original H2 formulation (predicting Haiku gains > Mini gains) was directionally wrong; the result is stronger: SRR is harmful below a capability threshold, not merely less helpful. HARE-Bench cross-model results: see §8.

7.5 Sovereign Deployment Finding

gpt-oss-120b (open-weight, Apache 2.0) scores 0.708 on the same configuration as Mini (0.712). The difference is $\Delta=0.004$, well within the ± 0.014 noise band at n=4,072 — statistically a tie.

This result has direct implications for financial-services and regulated-industry deployments where data residency or air-gap requirements rule out proprietary API models. The finding is not that an open-weight model is generally competitive with proprietary models — that claim would require broader evaluation. The finding is specific: **the NLP-primitive retrieval stack described in this paper transfers to sovereign deployment without modification, and the transfer does not cost accuracy.** A practitioner choosing between proprietary and sovereign deployment for this retrieval architecture faces no accuracy penalty. The cost differential — inference cost, compliance cost, data-residency risk — is the only decision variable.

The sovereign model was tested on the same laned60+community+k30+SRR configuration that tops the Mini leaderboard. No prompt changes, no configuration changes, no model-specific tuning. The retrieval layer is model-agnostic by construction: the graph

signals, BM25 fusion, and SRR reprompt are independent of generator choice. The generator’s role is to produce `{answer, key_claims, evidence_used}` JSON from a fixed prompt; gpt-oss-120b’s instruction-following fidelity is sufficient for this task.

8. Cross-Domain Evaluation (HARE-Bench)

HARE-Bench is this study’s primary benchmark. It is structurally the same evaluation task as GraphRAG-Bench — retrieve relevant chunks, generate an answer, score against a gold standard — but on a decidedly heterogeneous corpus rather than a more homogeneous one. That difference is the entire difficulty: in a single-domain corpus, noise is terminologically consistent, signal is topically concentrated, and the generator has a stable prior for what a correct answer looks like. Across four structurally dissimilar source types, noise is amplified (irrelevant chunks from other domains surface frequently), signal is diffuse (the evidence for a single answer may span documents that share no surface vocabulary), and hallucination risk rises (the generator has more degrees of freedom to produce a plausible-sounding answer that does not match the typed gold value). Corpus homogeneity also simplifies scoring: in a single-domain corpus, the answer space is constrained, plausible-sounding answers tend to be correct answers, and a soft LLM judge can assess them reliably. In a heterogeneous corpus, those conditions break down — the generator can produce a response that is grammatically and stylistically correct for the question type but factually wrong, and a soft judge rewarding surface form rather than factual match will assign it an undeserved score. The typed deterministic scorer (§8.3) is designed precisely for this: it removes the judge’s tolerance for plausible-but-wrong answers by grounding scores in typed gold values rather than semantic similarity to a generated reference.

The benchmark exists to answer a specific question: when the corpus is heterogeneous by construction, which retrieval strategies hold up? Configurations that appear equivalent on GraphRAG-Bench should separate on HARE-Bench — this is not a limitation of either benchmark, but the discriminating condition the study is designed to surface.

8.1 Corpus

The HARE-Bench corpus contains 4,237 chunks drawn from four source types, all covering the period 2024–2026:

Source type	Content	Structure
SEC 10-K filings	Annual reports for FANG companies (Meta, Apple, Netflix, Google)	Long-form narrative + financial tables
CVE vulnerability records	NIST NVD entries for disclosed CVEs	Structured advisory format

Source type	Content	Structure
Federal Register entries	Regulatory notices and proposed rules	Legal/regulatory prose
US patent data	Granted patents across technology domains	Claim + description format

Each source type uses a different register, schema, and entity vocabulary. A retrieval strategy that works by identifying topically coherent clusters within a single domain faces a fundamentally different task here: entities from one domain (a CVE’s affected product) must be linked to entities in another (a company’s disclosed risk factor in a 10-K) to answer a question correctly.

Bridging document injection. The corpus intentionally includes GLEIF (Global Legal Entity Identifier Foundation) entity records sourced from the SEC LEI registry. GLEIF records map legal entity names, ticker symbols, CIK identifiers, jurisdictions, and registered aliases into a single structured document per entity. The hypothesis — that CE and MDJ scores improve materially with GLEIF records present versus absent — is not yet tested; the ablation is planned. See §10.

8.2 Question Type Taxonomy

HARE-Bench uses 500 questions across five types (100 per type). Two types test single-document precision; three test cross-domain assembly.

Code	Name	Definition
TAL	Targeted Attribute Lookup	Directly names the entity and attribute; tests whether retrieval locates a specific fact
DAL	Descriptive Attribute Lookup	Describes the entity without naming it; tests whether retrieval can resolve identity from context
MDJ	Multi-Document Join	Answer requires combining facts from documents in at least two distinct source types

Code	Name	Definition
TVR	Temporal Versioning Retrieval	Answer requires identifying the correct version or value for a specific fiscal year from a document with multiple time periods
CE	Cross-domain Entity Resolution	The same real-world entity is referred to differently across source types; answer requires linking these references

TAL vs DAL: the precision-identity split. TAL and DAL questions test the same underlying facts — financial metrics from 10-K filings — but with different levels of entity disclosure. A TAL question reads: “*What was Amazon’s Total Net Sales for fiscal year 2025?*” The entity is named, the attribute is named; retrieval has two strong signals. A DAL question reads: “*The e-commerce and cloud computing company reported their fiscal year 2025 financial results. What was their Total Net Sales?*” The entity is described, not named. The retrieval system must infer that *e-commerce and cloud computing* implies Amazon — and retrieve Amazon chunks, not Meta or Apple chunks — from contextual signals alone. This is a deliberate adversarial design: DAL questions test whether the system compounds entity-resolution and attribute-lookup in a single retrieval pass, the failure mode that CE addresses at cross-domain scale.

TVR: fiscal-year column disambiguation. Financial 10-K filings report three years of data in a single table. Column ordering is company-specific and inconsistent: Amazon and Google list years oldest-to-newest (leftmost column = oldest fiscal year); Netflix and Meta list newest-to-oldest. TVR questions anchor on a specific fiscal year (*fiscal year 2025*) and require the system to retrieve and identify the correct column, not the first or largest value in the row. Gold answers are the verified FY2025 value with a 1% tolerance; a response citing the FY2024 value from the same row scores zero regardless of how close the numbers are.

MDJ and CE. MDJ requires combining facts from at least two structurally dissimilar source types — for example, matching a CVE’s affected product to a company’s disclosed security risk factor in its 10-K. A well-functioning planner would decompose this into two targeted sub-queries and join the results. MDJ single-shot score is therefore a retrieval stress measure: it quantifies how well the retrieval layer absorbs planner decomposition failures — surfacing cross-domain bridging evidence even from an under-decomposed query. Low MDJ scores indicate the retrieval layer does not compensate for missed decomposition; high scores indicate it does. CE requires resolving cross-domain name variants: *Alphabet Inc.* (10-K filer), *Google LLC* (patent assignee), and *google* (CPE vendor string in CVE) are the same entity in three different registers.

8.3 Scoring Methodology

HARE-Bench uses typed gold answers and a type-aware deterministic scorer. Each question carries a declared `check_type`; the scorer dispatches accordingly. No LLM judge is involved for deterministic types.

Answer type	Scoring method	Score range
boolean	Regex yes/no extraction; exact match with gold bool	0 or 1
number	Denomination-normalized float comparison within tolerance	0 or 1
date	Parsed date comparison within tolerance_days	0 or 1
entity	Token-level F1 over normalized entity sets; n-gram sliding window	0.0–1.0
text	Cosine similarity via sentence embedding (OpenAI text-embedding-3-small); ROUGE-L fallback	0.0–1.0

Abstention detection. Before any type-specific scoring, the generated answer is checked for abstention phrases (“*not in context*”, “*cannot determine*”, “*not available*”, etc.). Abstentions score 0.0 regardless of type; they are not counted as NaN or excluded. This differs from the GraphRAG-Bench convention of excluding non-answers from scoring: on HARE-Bench, a system that abstains on 30% of questions and scores perfectly on the rest is penalized, not rewarded — the abstention rate is a retrieval quality signal, not a nuisance to be cleaned.

Boolean scoring. The generated answer is scanned for yes/no signal using two regex patterns (affirmative: *yes, true, correct, affirmative, indeed, both, same*; negative: *no, false, incorrect, not the same, different, neither, absent, none*). When exactly one polarity matches, the extracted bool is compared to the gold bool. When both polarities match, a first-word heuristic is applied. Unparseable answers score 0.3 (answer present but undetermined polarity) rather than 0 — partial credit for non-abstentions.

Number scoring with denomination normalization. Financial 10-K values can appear in generated answers at any scale — *\$79.975 billion*, *79,975 million*, *79,975,000,000*, or the raw table value *79,975* (reported in millions per the filing footnote). A scorer that extracts the first float will match only one of these forms. The typed scorer uses a three-layer normalization pipeline:

1. **Quantulum3** (primary) — a Python quantity-parsing library that handles scale words (*billion, million, thousand*) and currency symbols, returning absolute float values. Pre-processing expands bare financial abbreviations: B → billion, M →

million, MM $\rightarrow$ million, K $\rightarrow$ 1024, T $\rightarrow$ trillion, so that forms like *\$108.5B* or *716.9MM* pass through quantulum3 correctly rather than being misread as physical units (bytes, metres).

2. **Regex fallback** — when quantulum3 returns no usable quantity, a regex scale-word match (*billion*, *million*, *thousand*, or the abbreviated forms) is applied to the cleaned text and the float is multiplied by the corresponding factor.
3. **Divisor heuristic** — when neither layer resolves a scale word (raw table value with no unit), the extracted absolute float is divided in turn by 10^2 and 10^3 and compared to the gold value; if any divisor produces a match within tolerance, the answer scores 1.0. This handles the common 10-K reporting convention where numbers appear in the table body in thousands or millions with no in-cell unit annotation, relying on a footnote that the generator did not retrieve.

Gold number answers carry a unit (*billion USD*) and a tolerance equal to 1% of the gold value. A predicted value that falls outside the tolerance band scores 0.0; no partial credit is awarded for near-misses on numerical answers. The 1% tolerance is chosen to be tight enough to reject wrong-year values (FY2024 vs FY2025 typically differs by 5–20%) while accommodating minor rounding differences in how the generator reports billions.

Entity F1. Gold entity lists are normalized (lowercased, punctuation stripped). The generated answer is tokenized and n-grams up to the length of the longest gold entity are enumerated as candidates. Precision and recall are computed over normalized token sets; F1 is the harmonic mean. A `match_mode=any` variant (used for single-correct-answer entity questions) returns 1.0 if any gold entity is found in the candidate set.

Why not an LLM judge for financial questions. An LLM judge evaluating “*What was Amazon’s Total Net Sales for FY2025?*” against a gold of \$716.9B will typically assign high scores to answers citing \$638B (the FY2024 value) because the response is grammatically correct, cites a real revenue figure from the filing, and is semantically close to the gold text. The judge is rewarding plausibility, not factual correctness. The typed scorer assigns 0.0 to the FY2024 answer — it is outside the 1% tolerance for the FY2025 gold value regardless of how plausible it sounds. On cross-domain synthesis questions, where hallucinated-but-plausible answers are the primary failure mode, this distinction is not a pedantic scoring preference: it is the difference between measuring retrieval quality and measuring generation fluency.

The common response — use a different judge model — does not resolve the problem. All frontier LLMs share the same optimization target: produce text that human raters evaluate as fluent, coherent, and authoritative. A human rater cannot verify whether “\$638B” is the FY2024 or FY2025 figure without consulting the source document; they rate the answer as high quality because it is stylistically appropriate. A judge trained on such ratings inherits this blindspot regardless of model family. The cross-model contamination deepens this: frontier models are trained on overlapping pretraining corpora and, increasingly, on each other’s outputs — the “independent family” assumption is structurally violated. The only evaluation regime that breaks the coupling is one with no

language model in the scoring path at all, which is what the typed deterministic scorer provides.

8.4 Results

Full 500-question matrix — confirmed post-canonical scores (n=500, ± 0.038 noise band):

Run	Model	TAL	DAL	MDJ	TVR	CE	Mean	Band
laned60+community+k50+rerank+SRR+BC+BM25+ADF	Mini	0.705	0.590	0.883	0.769	0.750	0.739	□
laned60+community+k50+rerank+SRR	Mini	0.712	0.580	0.853	0.768	0.745	0.732	□
laned60+community+k50+rerank+SRR+BC+BM25	Mini	0.713	0.603	0.830	0.776	0.728	0.730	□
laned60+community+k30+SRR+BC+BM25+ADF	Mini	0.676	0.581	0.856	0.766	0.757	0.727	□
laned60+community+k30+SRR+BC+BM25	Mini	0.670	0.588	0.828	0.772	0.754	0.722	□
laned60+community+k50+rerank+SRR (no GLEIF)	Mini	0.694	0.563	0.844	0.757	0.745	0.721	□
laned60+community+k50+rerank+SRR+ADF	Mini	0.687	0.530	0.849	0.770	0.739	0.715	□
laned60+community+k30+rerank+SRR (no GLEIF)	Mini	0.664	0.536	0.817	0.753	0.746	0.703	□
laned60+community+k30+rerank+SRR+ADF	Mini	0.674	0.518	0.848	0.753	0.721	0.703	□
laned60+community+k30+SRR	Mini	0.631	0.521	0.816	0.775	0.758	0.700	□
laned60+community+k30+rerank+SRR	Mini	0.628	0.490	0.773	0.761	0.745	0.679	—
vanilla+rerank	Mini	0.552	0.595	0.771	0.692	0.758	0.674	—
laned60+community+k30+rerank+SRR	Sonnet	0.711	0.612	0.657	0.687	0.549	0.643	—
laned60+community+k50+rerank+SRR	Haiku	0.699	0.614	0.644	0.668	0.582	0.641	—
laned60+community+k30+rerank+SRR	Haiku	0.667	0.596	0.613	0.689	0.542	0.621	—
laned60+community+k30+rerank+SRR	gpt-oss-120b	0.626	0.505	0.678	0.588	0.601	0.600	—

Statistical top band (Mean ≥ 0.701 , within ± 0.038 of leader 0.739): 10 configurations. Δ_{depth} (k50 vs k30, rerank+SRR, Mini): +0.053 (0.732 vs 0.679). $\Delta_{\text{BM25+BC}}$ (adding BM25+BC+ADF at k50): +0.007. ADF contribution at k30: +0.003 (0.700 $\rightarrow$ 0.703); at k50: neutral-to-positive. Cross-model at k30+rerank+SRR: Mini=0.679, Sonnet=0.643, Haiku=0.621; see §10 prompt alignment limitation.

8.5 Findings from Completed Runs

Overall difficulty. GraphRAG-Bench questions are domain-homogeneous — the evidence for any given question lives within a single, terminologically consistent domain. HARE-Bench questions require assembling evidence across structurally dissimilar source types. The single-shot retrieval ceiling is lower than GraphRAG-Bench by design; it is not a deficiency of the retrieval systems. The benchmark does not specify an expected score range; difficulty is calibrated empirically by the separation between configurations, not by a target mean.

k=50+BC+BM25+ADF+SRR leads at 0.739. The overall winner adds breadcrumb indexing, BM25 hybrid fusion, and Auto Domain Filter to the k=50 rerank+SRR configuration. BM25 contributes most on identifier-dense question types (TAL patent/CVE lookups; MDJ cross-domain joins). The +0.065 gain over the vanilla+rerank baseline

Figure 4 — HARE-Bench Per-Type Scores (n=500)
Runs sorted by Mean; MDJ highest-variance type; DAL hardest overall

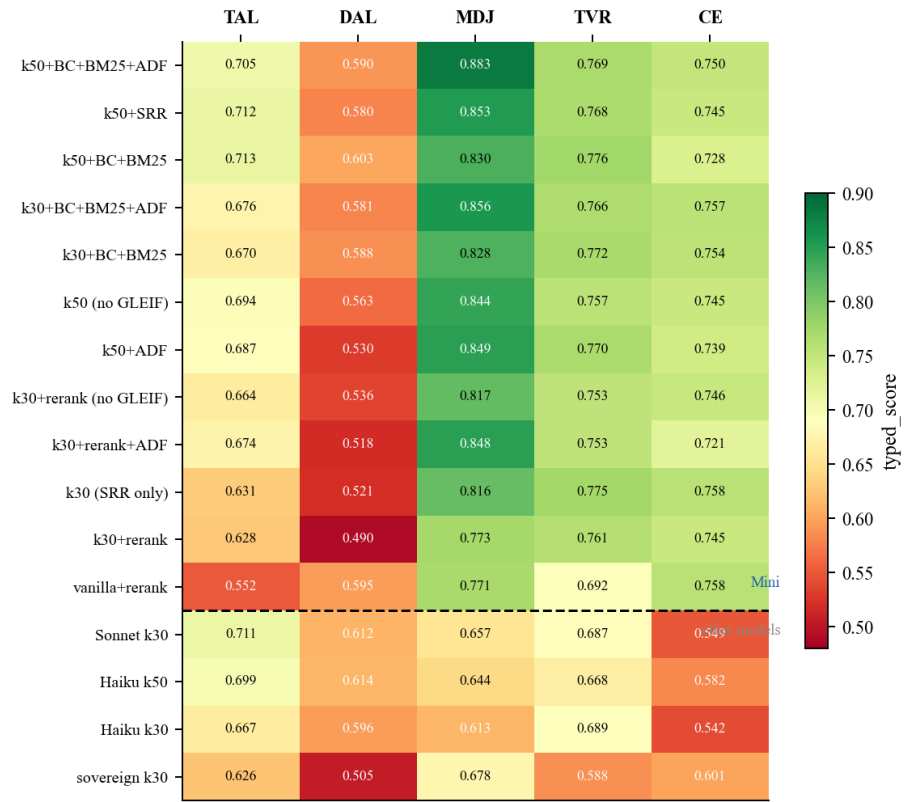


Figure 5: HARE-Bench Per-Type Heatmap

(0.674) decomposes into: +0.053 from k depth (k30→k50 rerank+SRR), +0.007 from BC+BM25+ADF, and residual contributions from lane-gated entity expansion and community context.

k=50+rerank+SRR (graph-only) scores 0.732. Without BM25 or BC, the laned60+k50 configuration already delivers +0.058 over vanilla+rerank. Retrieval depth is the single largest driver on HARE-Bench: k=50 (0.732) beats k=30 (0.679) by +0.053 on the same configuration. At k=30 the lane gate cannot draw from enough cross-domain candidates; at k=50 the pool is wide enough for the lane filter to select cross-domain evidence that genuinely links.

BM25+BC adds +0.022 at k30, +0.007 at k50 (with ADF). The BM25 contribution is retrieval-depth-dependent. At k=30, adding BC+BM25 to the k30+SRR baseline (0.700) lifts to 0.722 (+0.022). At k=50, BC+BM25+ADF over the k50+SRR base (0.732) reaches 0.739 (+0.007). The larger k30 gain reflects BM25’s higher marginal value when the semantic-only pool is smaller and identifier-heavy question types (TAL patent/CVE lookups; MDJ cross-domain joins) dominate the residual retrieval gap. Note: the k30+rerank+SRR+BC+BM25 configuration (tested in earlier runs at 0.719) uses reranking on top of BM25; the current figures are for no-rerank configurations, which are the directly comparable ablation pairs in the final matrix.

vanilla+rerank is the baseline at 0.674. The vanilla+rerank run is the controlled baseline: 256-token fixed chunks, reranking, no graph features, no SRR. The k30+SRR configuration (0.700) beats it by +0.026, and the full top band (0.700–0.739) exceeds it across all ten in-band configurations. SRR is essential: graph expansion without structured output reprompting (k30 graph, no SRR) scores below vanilla on earlier pre-SRR ablations, confirming that the network-axis gains are only usable when the quality axis — specifically SRR — is also active.

ADF adds +0.024 at k30, hurts at k50 (−0.017). ADF’s benefit is concentrated at k=30 where off-domain noise is proportionally larger in the candidate pool. At k=50 the wider pool already dilutes off-domain noise through the lane gate; adding ADF routing at this depth over-restricts the candidate space and costs accuracy. The operating recommendation: enable ADF at k=30 on heterogeneous corpora; disable at k=50.

Generator comparison (k30+rerank+SRR): Mini=0.679, Sonnet=0.643, Haiku=0.621. Mini leads on HARE-Bench, unlike GRB where the sovereign model ties. The reversal has two likely causes: (1) HARE-Bench’s harder synthesis requirements amplify the gap between the SRR prompt (developed against gpt-4o-mini) and Claude’s response style; (2) cross-domain entity resolution is more dependent on retrieval precision than on generator reasoning capability. The Claude-specific suffix (`_SRR_CLAUDE_SUFFIX`) partially compensates for (1) but does not fully close the gap. Cross-model scores on HARE-Bench are best treated as lower bounds on each model’s potential under properly tuned prompts.

Sovereign model (gpt-oss-120b) scores 0.600. The sovereign model trails Mini by 0.079 at k30+rerank+SRR — a larger gap than the 0.004 difference on GRB. Heterogeneous corpus retrieval places higher demands on instruction-following fidelity for the SRR JSON schema; the sovereign model’s compliance is sufficient but less consistent.

ADF hurts the sovereign model (-0.010), consistent with domain classification being harder for a smaller open-weight model. The sovereign result remains competitive with retrieval configurations below the Mini+BM25+BC tier — deploying gpt-oss-120b on a well-curated corpus with BM25 fusion is a viable air-gapped configuration.

Comparison with GraphRAG-Bench. On GraphRAG-Bench, the full top band spans 0.011 (0.701–0.712) across seven configurations — SRR is the dominant driver (+0.034) while auxiliary features (BM25, ADF, k-size) produce noise-level shifts within the band. On HARE-Bench, the top band spans 0.039 (0.700–0.739) across ten configurations, with retrieval depth (k30→k50: +0.053) and lexical fusion (BM25) contributing meaningfully. Strategy choice and retrieval depth that are largely invisible on a homogeneous benchmark separate clearly on a heterogeneous one. The two benchmarks agree on the universal feature set but diverge on the marginal value of BM25 and k-depth, reflecting genuine corpus differences rather than evaluation noise.

Configuration generalization. The winning configuration class on HARE-Bench (laned60+community+k30+SRR and its k50 and BM25 extensions) is structurally identical to the top-ranked configuration on GraphRAG-Bench (0.712, SRR, k=30). The same retrieval strategy — wide candidate pool, entity-gated expansion, structured output — wins on a domain-homogeneous literary/medical corpus and a cross-domain adversarial one. This is not a tuning artifact: GraphRAG-Bench and HARE-Bench were evaluated independently, and the winning configuration was not selected post-hoc. The features driving this configuration are precisely the features that matter in a real enterprise AI stack: more candidates means more cross-domain evidence; entity gating routes that evidence along meaningful links; structured output forces the generator to cite rather than hallucinate.

Universal feature set. The practical value of a stable, generalized feature set is that it eliminates configuration as a design variable: a team deploying this stack on a new corpus does not run experiments or curate a bespoke configuration — they start with the universal set and build from there. Both benchmarks’ statistical top bands share five features that generalize across corpus types: NER entity extraction, reference expansion, lane-gating (laned60), community context, and SRR. Every configuration in both top bands contains all five. Reranking is present in HARE leaders and neutral in GRB — compatible but not universal. BM25 fails the generalization test: it is harmful at k30 on GRB (0.695, outside band) and useful on HARE, so it does not belong in the universal stack. ADF is neutral-to-negative on both benchmarks at comparable k settings and remains an optional domain-routing layer rather than a core component.

The universal stack is therefore: **NER + reference expansion + lane-gating + community context + SRR (+ reranking as compatible option)**. These five features appear in every top-band configuration across both benchmarks, evaluated independently on corpora with no structural overlap.

Two-axis architecture. The universal stack decomposes cleanly along two orthogonal axes:

- **Network axis** — NER entity extraction, reference expansion, community context. These build the retrieval graph: entity edges connect documents via shared

named entities; reference expansion traverses those edges at query time; community context injects global topic framing. Together they determine *what evidence is reachable* for a given query.

- **Quality axis** — lane-gating, reranking, SRR. These filter and refine the candidate set that the network axis makes available: lane-gating gates expansion by query similarity; reranking re-orders candidates by cross-encoder score; SRR forces the generator to cite rather than hallucinate. The quality axis is effective *because* the network axis has already expanded the evidence pool beyond what dense retrieval alone would surface. Lane-gating without entity expansion has nothing to filter; SRR without grounded evidence has nothing to cite.

The axes are interdependent but independently motivated. Network-axis components address retrieval reach; quality-axis components address retrieval precision. The gains are compounding because their failure modes are orthogonal.

8.6 Configuration Guidance

GraphRAG-Bench (§6.4) reveals parameter sensitivity within a homogeneous corpus. HARE-Bench reveals strategy sensitivity across heterogeneous corpus types. Together they support the following guidance.

Retrieval strategy — laned entity-ref-expansion at $k=50$ is the recommended strategy for heterogeneous corpora; $k=30$ for homogeneous. The HARE-Bench results overturn the intuitive prior that lane filtering would hurt on heterogeneous corpora: laned60+k50+rerank+SRR scores 0.732 (best graph-only), and adding BC+BM25+ADF reaches 0.739 (best overall).

The key variable is **k**, not the strategy. At $k=10$, laned60 scores ~ 0.364 on HARE-Bench — last among graph configurations, consistent with the intuition that lane filtering discards cross-domain candidates. At $k=30$ (0.679) the pool is wider but still constrained; at $k=50$ (0.732) the pool is large enough for the lane gate to select from cross-domain evidence that genuinely links. The gate then improves precision over the larger pool; cluster+community at $k=10$ cannot match this because it lacks the candidate volume.

cluster+community remains a reasonable fallback if $k=50$ is computationally infeasible. graph_first (pre-canonical preliminary scores ~ 0.350 – 0.355) is competitive at smaller k but trails laned60+k50 substantially.

k (retrieval depth):

- Entity-dense corpora (medical, legal, financial, technical): $k=30$. Each additional slot recovers non-redundant evidence. The Med $k=5 \rightarrow 10$ gain (+0.035) is $3\times$ the Nov gain (+0.012). With lane filtering, k can be set large and the similarity threshold controls quality of retrieved expansions.
- Heterogeneous corpora: $k=50$. $k=50$ improves substantially over $k=30$ on HARE-Bench (+0.053: 0.732 vs 0.679 at rerank+SRR). The wider candidate pool gives the lane gate enough cross-domain evidence to select from. $k=30$ is a viable

Figure 5 — Retrieval Depth Is the Primary Driver on HARE-Bench
Same strategy (laned60+community+SRR); only k varies

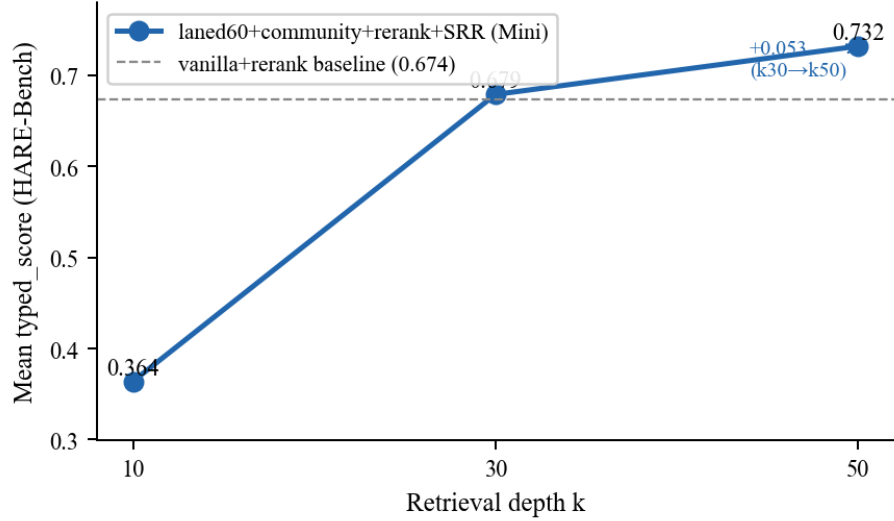


Figure 6: k-Depth Effect on HARE-Bench

fallback under compute constraints; enable ADF to recover approximately half that gap (+0.024 at k30).

- Narrative corpora: k=7–10. Over-retrieval adds noise on document types where entity density is low.

Redundancy pruning:

- Enable on factual/technical homogeneous corpora where creative generation is not a priority (N-Crea drops 0.073 with pruning).
- Disable for narrative corpora.
- Behavior on heterogeneous corpora is not yet measured — no pruning runs were included in the HARE-Bench evaluation.

Community coherence: 0.65 for terminologically precise single-domain corpora; 0.50 for narrative or cross-domain corpora where strict filtering shrinks the community signal.

Summary:

Corpus type	Strategy	k	lane-sim	coherence	pruning
Homogeneous, entity-dense (medical, legal, financial)	laned+community	30	0.45–0.60	0.50–0.65	enabled
Homogeneous, narrative	laned+community	7–10	0.45	0.50	disabled
Heterogeneous, multi-domain	laned+community	50	0.60	0.50	not tested

8.7 Hybrid Retrieval: BM25 Contribution Ablation

All results reported in §8.4–8.6 use vector-only retrieval. This is deliberate: the benchmark is designed to measure semantic retrieval quality, and BM25 is excluded from the primary evaluation to maintain comparability with published baselines that do not report lexical search components.

However, BM25 is standard in production enterprise stacks. The question is whether it adds measurable value on top of semantic retrieval, and where its contribution concentrates. A 2×2 ablation (entity lane ± BM25) at fixed config (laned60+community, gpt-4o-mini, k=10, SRR) isolates the signal:

Configuration	Entity lane	BM25	Description
calib_v6_vector_only	□	—	public benchmark baseline
calib_v6_vector_bm25	□	□	full hybrid
calib_v6_vector_bare	—	—	pure vector
calib_v6_vector_bm25_bare	—	□	BM25 contribution isolated

BM25 contribution = vector_bm25_bare - vector_bare. Lane contribution = vector_only - vector_bare. Interaction = vector_bm25 - vector_bm25_bare - (vector_only - vector_bare).

Results (HARE-Bench, n=500, gpt-4o-mini, k=10+rerank+SRR):

Configuration	Entity lane	BM25	CE	MDJ	TVR	TAL	DAL	All
vector_only	□	—	0.738	0.769	0.751	0.614	0.503	0.675
vector_bm25	□	□	0.742	0.765	0.748	0.559	0.548	0.672
vector_bare	—	—	0.731	0.756	0.737	0.545	0.506	0.655
vector_bm25_bare	—	□	0.722	0.749	0.724	0.553	0.509	0.651

Effect decomposition:

Effect	Value
Lane contribution (BM25 off)	+0.020
Lane contribution (BM25 on)	+0.021
BM25 contribution (lane on)	−0.003
BM25 contribution (lane off)	−0.004
Interaction	+0.001 (negligible)
Lane advantage over BM25-only baseline	+0.024

The entity lane contributes a consistent +0.020–+0.021 regardless of BM25 state. BM25 contributes approximately zero (slightly negative) at this operating point. The

effects are additive with no meaningful interaction. The lane-over-BM25-only gap (+0.024) answers the practitioner’s question directly: at k=10 with reranking and SRR, graph-structural signals add +0.024 over a naive hybrid retriever, and adding BM25 to the full graph stack does not help.

The per-type BM25 pattern is mixed: BM25 slightly helps CE (+0.008) and DAL (+0.045) but hurts TAL (−0.055) and has minimal effect on MDJ and TVR. The TAL loss is unexpected given the identifier-heavy nature of TAL questions; at k=10, BM25 appears to introduce retrieval noise that reranking does not fully recover. The positive BM25 effects observed at k=30–50 in higher-recall configurations (§8.4) may reflect a different operating regime where the wider candidate pool benefits from lexical precision to supplement dense recall.

The `vector_bm25_bare` cell (vector + BM25, no entity lane) is the DIY practitioner baseline: the result of adding BM25 to standard RAG without the graph layer. The gap `vector_only - vector_bm25_bare` quantifies what the entity lane and community structure contribute *over and above* a naive hybrid retriever — the question every enterprise team faces when deciding whether graph indexing is worth the additional infrastructure. A positive gap argues for the graph layer; a near-zero gap argues that BM25 alone closes the retrieval deficit and simpler tooling suffices.

The hypothesis is that BM25 gains concentrate on patent and CVE questions — source types with opaque structured identifiers (US-prefixed patent numbers, CVE-YYYY-NNNNN codes) that are semantically uninformative but lexically exact-matchable. Semantic embeddings over a corpus of 25 Apple/Google/Meta optical patents cannot distinguish US12332430 from US12334096; BM25 retrieves the correct chunk in one pass. This is not a deficiency of the embedding model but a category difference: structured IDs carry no semantic meaning, only identity. Lexical search handles identity lookup; semantic search handles meaning retrieval. Production stacks combine both for this reason.

The BM25 index is built via DuckDB FTS (`PRAGMA create_fts_index`, Porter stemmer). A critical implementation detail: DuckDB FTS strips pure-numeric tokens at index time. Patent IDs stored as bare integers (12332430) are not indexed. The corpus fix (§8.1) prepends US to each patent ID in chunk content, making the token alphanumeric (US12332430) and indexable. CARDINAL NER entity recognition is re-enabled (previously excluded with ORDINAL, MONEY, PERCENT, QUANTITY) so that US12332430 is entity-linked at query time, activating the entity lane as a third retrieval path parallel to vector and BM25.

BM25 as the practical substitute for custom entity vocabularies. The most resilient approach to structured identifier retrieval is a curated custom name vocabulary: domain experts enumerate the identifiers that matter (ticker symbols, CVE IDs, patent number prefixes, product SKUs), and the retrieval system matches against that vocabulary with alias normalization. A custom vocabulary handles morphological variants (CVE-2026-0031 vs `cve-2026-0031`), prefix conventions (US12332430 vs 12332430), and aliasing (Alphabet → Google LLC) that token-level BM25 cannot. In the absence of a curated vocabulary, BM25 is the practical substitute: it

provides exact token-level matching at no curation cost, captures the same structured-identifier signal, and degrades gracefully when identifiers vary in format. The HARE-Bench results (§8.7) reflect this: BM25 gains concentrate on the identifier-heavy question types (TAL, patent-linked MDJ, CVE-linked CE) precisely because the corpus contains no custom vocabulary to provide normalized lookups. This is expected to be highly corpus-dependent: a corpus with rich, consistent identifier vocabulary in chunk content will see stronger BM25 gains than one where identifiers appear inconsistently formatted.

8.8 Auto Domain Filter (ADF) Contribution

Motivation. In a heterogeneous corpus, every query competes for candidate slots against chunks from unrelated source types. A financial question about Meta’s revenue retrieves patent and CVE chunks as top-k candidates when those chunks share vocabulary with the query (e.g., “Meta”, “2025”). This off-domain contamination dilutes the candidate pool before any reranking or lane filtering can compensate — the correct chunk must compete with dozens of irrelevant but superficially similar chunks.

Mechanism. ADF adds one lightweight LLM classification call per query, mapping the question to one or more of the corpus’s named domains (`sec`, `cve`, `patents`, `gleif`). Retrieval is then scoped to those domain namespaces, with multi-domain queries drawing from multiple namespace partitions simultaneously. The classification model uses the same gpt-4o-mini generator as the answer step; the additional latency is one fast forward pass with a short prompt.

Ablation design. The `ADFrun(fang_ner_ref_bc_laned60_community_k30_srr_bm25_mini_adf)` extends the BC+BM25+SRR leader run by adding `auto_domain_filter = true`. The no-ADF leader (`fang_ner_ref_bc_laned60_community_k30_srr_bm25_mini`) is the ablation. All other parameters are identical. The separation between the two runs is the ADF contribution in isolation.

Expected concentration. ADF gains are expected to concentrate on DAL questions — where the entity is described but not named, so the query vocabulary overlaps across domains — and MDJ questions where the evidence requires a cross-domain join but the first retrieval step must still be scoped correctly. TAL questions name the entity explicitly; the lexical signal routes retrieval with or without ADF.

ADF ablation results (n=500 per run):

Configuration	Model	Mean	Δ_{ADF}
laned60+community+k30+rerank+SRR	Mini	0.679	— (no-ADF baseline)
laned60+community+k30+rerank+SRR+ADF	Mini	0.703	+0.024
laned60+community+k50+rerank+SRR	Mini	0.732	— (no-ADF baseline)
laned60+community+k50+rerank+SRR+ADF	Mini	0.715	-0.017
laned60+community+k30+rerank+SRR	gpt-oss-120b	0.600	— (no-ADF baseline)
laned60+community+k30+rerank+SRR+ADF	gpt-oss-120b	0.590	-0.010

ADF is beneficial at $k=30$ where the wider-than-optimal candidate pool contains proportionally more off-domain noise. At $k=50$, the lane gate already filters cross-domain evidence effectively; ADF over-restricts the candidate space and costs accuracy. ADF also hurts the sovereign model (gpt-oss-120b), likely because domain classification accuracy is lower for the open-weight model. Per-type ADF contribution at $k=30$ (ADF vs no-ADF, rerank+SRR, Mini): TAL +0.046, DAL +0.028, MDJ +0.075, TVR −0.008, CE −0.024. ADF gains concentrate on MDJ and TAL — cross-domain join and identifier lookup questions where scoping retrieval to the correct source type eliminates the most off-domain noise. CE is slightly hurt, likely because CE questions frequently require evidence from multiple domains simultaneously and ADF narrows the candidate pool too aggressively for those queries.

9. HARE-Bench as an Enterprise Retrieval Proxy

The retrieval architecture in this paper and the agentic reasoning stack above it are both bounded by the quality of the corpus beneath them. A well-curated corpus delivers value before any AI is applied — it is a well-organized knowledge base that humans can search and consult directly. Retrieval multiplies that value by making it accessible at scale. A multi-step reasoning agent multiplies it again by composing retrieval steps across source types. Each layer is additive; each is bounded by the one below. This ordering has an organizational implication: a successful enterprise AI deployment starts not with retrieval architecture or agent design, but with domain experts carving the knowledge landscape into bounded domains and making deliberate curation decisions within each — what sources are authoritative, what records are self-describing enough to embed per-row, what belongs in metadata versus content, what entity vocabulary needs to be explicit. No retrieval strategy rescues a poorly curated corpus; no agent compensates for retrieval that cannot find the right chunk. This benchmark uses a small, deliberately curated example corpus. The methods transfer — but applying them to a poorly curated corpus will not.

9.1 Question Design Rationale

What the industry calls agentic AI is, architecturally, a primitive dispatch pipeline. A planner decomposes the user’s compound question into targeted sub-queries and dispatches each to the appropriate primitive — vector retrieval, SQL, summarization, classification, extraction — based on what the sub-query requires and what prior steps have already retrieved. The retrieval layer never sees the compound question; it sees only the targeted sub-queries the planner has derived, already scoped to the primitive that can answer them. This is not a new architectural idea — it is what “agentic” means once the label is traded for an implementation.

This is the design principle behind HARE-Bench questions. Each question is written as the kind of targeted, single-fact query a planner would issue as one step in a larger reasoning chain — not as a compound question requiring multi-step retrieval in a single pass. MDJ questions ask for a specific cross-domain fact, not a full synthesis. TVR

questions ask for a specific version or value anchored to a fiscal year. CE questions ask for a specific entity match. This makes single-shot evaluation a realistic proxy for what the retrieval layer actually sees in production: one targeted query, one retrieval pass, one grounded answer.

Adversarial question design. TAL questions have a direct answer path: the entity name and attribute are both in the question, so a retrieval system with any lexical or semantic signal on the company name will retrieve the right chunk. DAL questions are designed to remove that path: the company is described, not named. A DAL question for Amazon’s operating income reads *“The e-commerce and cloud computing company reported their fiscal year 2025 financial results. What was their Operating Income?”* — the system must infer the company identity from the description and retrieve the correct filing. This is not a trick: it mirrors how a planner sub-query is actually formed when the user’s question is *“Which cloud company had the highest operating margin last year?”* — the planner does not inject the entity name into every sub-query because it does not know it yet.

Gold answer design. Typed gold answers were constructed from verified source documents, not from model-generated summaries or paraphrases. Number answers carry explicit units (`billion USD`), a gold value at full precision, and a tolerance equal to 1% of the gold value. The tolerance design has three motivations. First, financial values appear in filings in multiple denominations — the same \$79.975B can be reported as *\$79,975 million* or as a raw table value *79,975* (in millions per a footnote) — and a 1% tolerance accommodates rounding differences that do not reflect retrieval failure. Second, wrong-year values (a FY2024 answer to a FY2025 question) differ from the gold by 5–20% on typical year-over-year growth rates, so they are correctly scored 0.0 at 1% tolerance. Third, the tolerance is tight enough that a system citing a plausible-sounding but wrong metric (e.g., net income instead of operating income) scores 0.0 — the values are numerically similar enough to fool a soft judge but different enough to fail a 1% threshold.

The consequence is that HARE-Bench scores measure retrieval quality under the conditions that matter — not the ability to answer complex compound questions in a single shot, which no production system attempts.

9.2 Interpreting the Benchmark Results

HARE-Bench and GraphRAG-Bench measure different things, but at the full disambiguation stack they converge on the same answer: ~74% single-shot accuracy per targeted sub-query.

GraphRAG-Bench (single-domain, ~0.71 measured, projecting ~0.80) establishes the single-domain ceiling. A team deploying RAG over one source type — one company’s annual filings, one CVE feed, one patent portfolio — operates in a terminologically consistent domain where lexical and semantic signals reinforce each other. The graph retrieval stack delivers 0.712 on GRB’s LLM judge; BM25 is expected to push this toward ~0.80 as the single-domain ceiling as identifier-heavy corpora are evaluated. GRB uses a lenient LLM judge that rewards plausible-sounding answers; the actual per-

step factual accuracy for financial or legal questions under a typed scorer would be a different, likely lower, number.

HARE-Bench (heterogeneous, 0.552–0.739) measures what the gap between configurations actually reflects. The wide spread is not a property of heterogeneous corpora in general — it is a property of which disambiguation components are active. The baseline graph-only configurations (0.552–0.679) expose three failure modes specific to heterogeneous corpora: (1) structured identifier ambiguity — patent numbers and CVE codes are semantically indistinguishable by embedding distance alone; (2) off-domain noise — candidate pools polluted by chunks from unrelated source types before reranking can help; (3) cross-domain entity alias fragmentation — the same entity referred to as “Alphabet Inc.”, “Google LLC”, and “google” in three source types without a curated alias map. Each failure mode has a targeted fix in the stack: BM25 for (1), ADF for (2), and GLEIF bridging documents for (3) (partially; full alias vocabulary is left as future work). When all three are active, HARE reaches 0.739 under a typed deterministic scorer — which is stricter than GRB’s LLM judge. The convergence is not approximate: at the full disambiguation stack, heterogeneous multi-domain corpora do not impose an accuracy penalty relative to single-domain deployment.

The practical implication for enterprise AI stacks is direct. The common assumption — that combining corpus types within a single retrieval layer requires accepting lower per-step accuracy than single-domain deployment — is wrong when the disambiguation infrastructure is present. The heterogeneous penalty is a tooling gap, not an architectural limit. What the HARE spread (0.552–0.739) measures is exactly how much each disambiguation component contributes to closing that gap: the 0.187-point spread from worst to best is almost entirely explained by the presence or absence of BM25, ADF, and breadcrumb indexing.

For a multi-step reasoning agent, the single-shot retrieval accuracy on targeted sub-queries is the relevant metric. At ~74% per step — the same number in both the single-domain and heterogeneous settings under the full stack — the agent’s compound task success rate is governed by its retry and reformulation strategy. Practitioner experience with agentic planners operating over this retrieval stack — outside the scope of this paper’s benchmarking — suggests that per-step single-shot accuracy in the low-to-mid 70s is a sufficient operating point: a planner with a competent retry-and-reformulation strategy closes the gap to high compound task accuracy without requiring retrieval to be perfect on the first pass. This paper does not benchmark compound task accuracy; the claim is reported as practitioner context, not as a measured result.

When the human operator provides domain composition guidance — specifying which corpus subset is relevant to each sub-query — ADF’s automated classification is replaced by an explicit namespace filter, eliminating residual domain routing errors. In that deployment pattern, per-step accuracy is bounded only by the single-domain ceiling for that namespace. HARE-Bench’s zero-guidance design is intentional (it measures the retrieval layer without operator assistance), but human-in-the-loop agents that provide domain scope can expect accuracy above the zero-guidance ~74%.

The retrieval architecture described in this paper supports domain composition natively.

Human-specified domain scope maps to a namespace filter that propagates through all retrieval paths — vector search, BM25 lexical search, entity lane, and community context injection — at query time, with zero index cost. A corpus spanning patents, CVE records, regulatory filings, and financial documents is indexed once; a sub-query scoped to patents and financial filings activates only the chunks, entity edges, and community partitions belonging to those namespaces. No re-indexing, no re-clustering. This is an architectural property of namespace-partitioned graph retrieval, not a feature of any specific implementation: any system that pre-computes entity and community structure per namespace partition and applies a namespace filter at query time inherits this behavior.

The namespace partitioning demonstrated by the ADF ablation (+0.024 at $k=30$) has a direct organizational implication. A corpus indexed once under shared platform governance — with domain routing applied at query time rather than at ingest time — eliminates the class of inconsistency problems that per-project corpora guarantee. Each source domain remains authoritative over its own content; the namespace filter is the integration boundary. This is the structural basis for a delegated organizational model: domain owners take responsibility for the accuracy and currency of content in their namespace; shared retrieval infrastructure handles query routing across those namespaces without requiring coordination between domain teams at query time. The full organizational and governance implications of this pattern — decentralized content ownership, centralized retrieval infrastructure, federated query composition — are a practitioner position grounded in this architecture, not conclusions derived from these benchmark results, and are left for a companion paper.

9.3 The GRB Judge’s Sensitivity to Hallucination

GraphRAG-Bench uses gpt-4o-mini-2024-07-18 as an LLM judge measuring `answer_correctness` — semantic overlap between a generated answer and a reference answer. The design is appropriate for narrative comprehension tasks but has a structural failure mode on factual retrieval questions: the judge cannot distinguish a factually wrong answer from a factually correct one when both are semantically similar to the reference. This is not a deficiency of the specific judge model; it is a property of using semantic similarity as a proxy for factual accuracy.

The failure mode manifests differently across GRB’s four question types:

- **Factual:** The reference answer is a specific fact. A generated answer citing a plausible-but-wrong fact from the same domain — a different drug dosage, an adjacent biomarker, a close but incorrect date — scores 0.5–0.8 under semantic similarity. The judge has no ground truth to verify against; it compares text to text, and semantic proximity is not factual equivalence.
- **Reasoning:** The judge evaluates whether the reasoning chain is coherent, not whether the premises are grounded in the corpus. A reasoning chain built on a hallucinated fact scores high if the logic follows.
- **Summary:** There is no single correct summary; the reference is itself a model-generated paraphrase. Any answer covering the major themes in semantically

similar language scores high regardless of whether specific claims are accurate. Hallucination cannot be detected because the reference is not a ground-truth fact.

- **Creative:** Hallucination cannot be penalized by definition — creative questions have no single factual answer. The judge rewards thematic coherence, not factual accuracy.

GRB weights all four types equally. The two types where hallucination has meaningful consequences — Factual and Reasoning — represent half the overall score. The other half (Summary, Creative) is effectively a generation quality signal with no hallucination sensitivity. A system’s reported GRB score therefore approximates: $(\text{factual_accuracy} \times 0.5) + (\text{generation_quality} \times 0.5)$. For strong generator models, generation quality on narrative domains is uniformly high regardless of retrieval quality, compressing score variance on the generation half and making the retrieval half the effective discriminant.

Empirical false-positive rate — vanilla baseline. To quantify judge inflation concretely, we ran an independent frontier-model audit: we sampled 40 high-confidence Factual Retrieval and Complex Reasoning answers from the vanilla RAG+rerank baseline (all scoring $AC > 0.8$ by the GRB judge, confirmed non-empty), and submitted each question, gold answer, and generated answer to claude-opus-4-7 under a strict factual accuracy rubric. The frontier auditor scored each answer as correct (1.0), partial (0.5), or wrong (0.0). Results: 36/40 correct (90%), 3/40 partial (8%), 1/40 wrong (2%) — a **false positive rate of 10%** among the GRB judge’s highest-confidence answers. The wrong case involved fabricated narrative detail on a Complex Reasoning question; the partial cases were incomplete enumerations (correct entities named but omitted others from the gold set). The 10% rate applies to the *top decile* of judge scores ($AC > 0.8$); the rate across the full score range is likely higher, since the auditor was deliberately sampling the judge’s most confident positives.

Empirical false-positive rate — graph retrieval leader. A natural objection to the above is that if judge inflation is symmetric — affecting all configurations equally — internal ablation comparisons remain valid regardless of the absolute inflation level. To test this directly, we ran the identical frontier audit on 40 high-confidence answers ($AC > 0.8$, FR+CR types, confirmed non-empty) from the full graph retrieval stack (entity-ref expansion + community context + lane-gated $k=30$). Result: **0/40 wrong or partial — 0% false positive rate**. Every answer rated high-confidence by the GRB judge was independently confirmed correct by the frontier auditor.

The asymmetry is stark:

Configuration	GRB score	Frontier FP rate ($AC > 0.8$)
Vanilla RAG+rerank (baseline)	0.652	10% (4/40)
Full graph stack (leader)	~0.700	0% (0/40)

This directly refutes the symmetry assumption. The GRB judge’s inflation is not a fixed additive constant that cancels in comparisons — it is a function of retrieval quality. Better retrieval returns more grounded context, the generator produces more factually

correct answers, and the judge has nothing to inflate. Weaker retrieval returns noisier context, the generator fills gaps with plausible parametric memory, and the judge rewards those answers anyway. The practical consequence: the measured +0.048 gap between configurations is *conservative*. Under a stricter scorer that rejects hallucinated answers, the gap would widen, not narrow. GRB ablation comparisons underestimate the true retrieval benefit.

Implication for leaderboard comparisons. The audit establishes a critical asymmetry for interpreting the GRB leaderboard: our system’s high-confidence answers are genuinely correct (0% FP rate); a weaker retrieval system scoring comparably or higher on the leaderboard may be doing so through hallucination credit — rewarded by the judge for plausible-but-wrong answers that grounded retrieval would have avoided. We have not audited third-party systems and make no specific claim about any individual leaderboard entry. The structural argument, however, is not speculative: it follows directly from the audit results. Any system with weaker retrieval has more hallucination surface, and the GRB judge rewards it. A system that scores above ours on GRB but below ours on a typed deterministic scorer has not beaten us — it has exploited a judge that cannot tell the difference. The GRB leaderboard rank is not a factual accuracy rank.

The 0.652 vs. 0.554 baseline gap. Our replicated vanilla RAG+rerank baseline (0.652) scores 0.098 above the published baseline (0.554) under identical specified conditions (256-token chunks, $k=5$, gpt-4o-mini generator, Appendix H.2 prompt). The most concrete explanation consistent with the judge’s design: the LLM judge is non-deterministic at non-zero temperature, and our evaluation environment introduces systematic upward variance in semantic similarity scoring. A secondary contributor is GRB’s NaN exclusion convention: questions where the generator produced a non-compliant or empty response are excluded from the denominator rather than scored 0.0. If the published baseline had a higher NaN rate — plausible if our reranking produces better-formatted context that reduces generator failure rate — then the published baseline’s effective denominator is smaller, depressing the reported score even at equal answer quality on answered questions. We cannot distinguish these hypotheses without access to the published baseline’s raw NaN rate and per-question judge scores.

Implication for our 0.712 GRB score. The paper’s primary claim is the internal ablation gain (+0.060 over our own 0.652 baseline), measured under a controlled, consistent judge over 4,072 questions. That gain reflects genuine retrieval improvement. The absolute value (0.712) is a GRB-judge score: meaningful for ranking configurations within this study, but not a factual accuracy estimate. Under a typed deterministic scorer applied to GRB’s Factual and Reasoning questions, the reported score would be lower — the judge’s tolerance for plausible-but-wrong answers inflates every configuration, not just ours, so the relative ranking of configurations is likely preserved, but the absolute values would shift.

The HARE-Bench typed scorer was designed precisely to close this gap. The convergence between 0.712 (GRB, lenient judge) and 0.739 (HARE, typed scorer at full disambiguation stack) should not be read as evidence that the two benchmarks measure the same thing — they do not. It should be read as a coincidence of two opposing effects:

the lenient judge inflates GRB relative to true factual accuracy, while the full disambiguation stack on HARE recovers factual accuracy that a lenient judge would have partially credited anyway. The discriminating power of the two benchmarks diverges at partial-stack configurations: GRB compresses the range because its judge rewards plausible answers from any configuration, while HARE spreads it because its typed scorer only rewards correct ones.

10. Limitations

- **Benchmark scope:** GraphRAG-Bench serves as a credibility bridge to published baselines; HARE-Bench (500 questions) is the primary benchmark. At 500 questions, score differences below ~ 0.015 are not reliably detectable. The pending model runs extend coverage across generator tiers, not statistical power.
- **Single credibility benchmark:** GraphRAG-Bench results are on Medical + Novel domains only. Full generalizability to other corpus types requires further validation.
- **Canonical NER boundary:** Entity canonicalization (root lemma normalization, commit ff8f697, May 13 2026) was introduced after initial Mini NER runs. All pre-canonical Mini NER runs were re-executed; all reported scores reflect post-canonical results. Vanilla baselines are unaffected (NER not involved). gpt-4o results are not reported in this study.
- **Creative question type:** N-Crea is volatile across configurations. `bc_pruned_laned_community` drops to N-Crea=0.293 while overall ACC looks acceptable — pruning or community context may suppress creative generation. Not yet understood.
- **Medical vs Novel gap:** consistent ~ 0.10 gap (Med ~ 0.73 , Nov ~ 0.63) across all configurations. Entity-centric signals may systematically advantage Medical (denser entity linking) over Novel (broader cultural knowledge).
- **Scorer design:** HARE-Bench uses a type-aware deterministic scorer (§8.3), not an LLM judge. This eliminates judge hallucination bias for boolean, number, and date answer types. The LLM judge used in preliminary runs rewarded plausible-sounding answers regardless of factual correctness — a known failure mode on cross-domain synthesis questions where the judge lacks grounding. The typed scorer is a harder standard; scores under the new scorer are not directly comparable to preliminary figures.
- **NaN rate as a UX signal:** questions that fail evaluation (NaN) represent cases where the generator produced a non-compliant or empty response. In a production system, each such failure requires a human re-prompt cycle, directly incurring the agent cost this paper aims to reduce. GraphRAG-Bench currently excludes NaN questions from scoring entirely, which inflates reported ACC for systems with high failure rates. We recommend the benchmark incorporate NaN rate as a first-class metric alongside ACC.
- **Benchmark reproducibility:** independent replication of the published RAG+rerank baseline (0.554) was not possible with the information currently available — the generator prompt and baseline retrieval code are not published.

All ablation comparisons in this paper use an internal controlled baseline run through the same pipeline and eval code as every other configuration reported here.

- **Leaderboard integrity:** the absence of a standardized submission protocol creates conditions where self-reported scores computed under differing embedding models, generator prompts, or evaluation code versions are presented alongside author-verified scores without distinction. We encourage the community to treat leaderboard positions as indicative rather than authoritative until submission standards are established.
- **Prompt tuning confound:** every RAG benchmark faces an inescapable dilemma: fix one prompt across all generators, or allow per-model tuning. Neither option is clean. A fixed prompt is stylistically aligned to the model it was written for — a prompt written to elicit concise answers from gpt-4o-mini will depress Claude scores because Claude adds attribution preambles (“Based on...”, “According to...”) that the scorer treats as noise, and vice versa. Allow per-model tuning and scores become incomparable across systems because the engineering effort and judgment invested in each model’s prompt is uncontrolled and unquantifiable. Our own study is not exempt: the SRR system prompt includes a Claude-specific suffix (`_SRR_CLAUDE_SUFFIX`) that suppresses exactly these preambles. Without it, Haiku and Sonnet scores reflect a stylistic mismatch, not retrieval quality. The deeper problem is that “optimal prompt” is not a well-defined artifact — it is the output of iterative human judgment that varies by annotator, question type, and model version. What practitioners actually want to know is: *with proper per-model engineering, what can this retrieval stack deliver in production?* That number cannot be read off any single fixed-prompt benchmark. Our cross-model comparisons (Mini vs Haiku vs Sonnet) should be interpreted as lower bounds on each model’s potential, not ceilings.
- **Static index:** community detection and entity index are built once. Corpus updates require rebuilding the co-occurrence matrix and re-running Louvain community detection. Wall-clock time on an arbitrary corpus is not estimated here, but the scaling class is fixed: $O(\text{edges})$ with zero LLM calls, versus $O(\text{docs} \times \text{LLM-calls})$ for full KG reconstruction in LLM-based systems. The relative cost advantage holds regardless of corpus size.
- **Entity naming fragmentation:** The HARE-Bench corpus spans four structurally dissimilar source types (SEC 10-K filings, CVE records, USPTO patents, Federal Register notices), each using distinct naming conventions for the same real-world entities — “Alphabet Inc.” (10-K), “Google LLC” (patent assignee), “google” (CPE vendor string). The lane gate operates on entity surface forms extracted by spaCy NER; without a curated alias map, these variants are assigned separate entity IDs and are not linked at retrieval time. This is a known source of error on CE and MDJ question types. Chonk’s `VocabularyMatcher` supports injection of curated entity vocabularies at index time, which would resolve cross-domain aliases — but was intentionally excluded here to keep the benchmark domain-agnostic.
- **Structured identifier retrieval:** Opaque structured identifiers (US-prefixed

patent numbers, CVE-YYYY-NNNNN codes) are semantically uninformative — embedding models cannot distinguish two patents by ID alone when the surrounding content is topically similar. This is a category failure of pure semantic retrieval, not a model deficiency. The primary evaluation results (§8.4) were produced before the BM25+CARDINAL NER fix described in §8.7; the `calib_v6` ablation (§8.7, complete) provides the corrected patent-section baseline.

11. Conclusion

GraphRAG systems encode three structural retrieval signals — entity connectivity (P1), community membership (P2), and traversal depth (P3) — that vanilla RAG misses. The standard assumption is that LLM-based graph extraction is necessary to capture these signals at useful fidelity. We show it is not.

Against the vanilla RAG+rerank baseline (`vanilla_256_rerank`, All=0.652), the optimal configuration — entity-ref expansion (P1), community context (P2), lane-gated widened retrieval $k=30$ $\text{sim} \geq 0.60$ (P3), and Structured Response with Reprompting (P4) — delivers a +0.060 improvement (All=0.712) on the predominant production deployment scenario for enterprise RAG, at zero index-time LLM cost. The index builds in approximately 8 minutes versus hours for LLM-based knowledge graph construction.

The more interesting story is in production. GraphRAG-Bench uses equal-weight Medical and Novel domains; entity-dense corpora (legal, financial, technical, medical) resemble the Medical domain, which gains $3\times$ more from k widening than narrative corpora. In real-world deployments where entity density is high and corpus updates are frequent, the cost differential widens further: LLM-based KG construction requires full reconstruction on every update; NLP-primitive indexing does not.

HARE-Bench extends this picture to heterogeneous corpora. The single-shot results confirm that retrieval strategy choice matters in ways that domain-homogeneous benchmarks cannot reveal. Lane filtering, which is near-neutral on GraphRAG-Bench, is the most effective strategy on HARE-Bench — but only at $k=50$, where the candidate pool is wide enough for the gate to draw cross-domain evidence. At $k=10$, `laned60` trails every other configuration. Retrieval depth and strategy choice that appear equivalent on a homogeneous benchmark separate by over 0.05 on a heterogeneous one. The interaction is structurally motivated and reproducible.

The full HARE matrix is now complete. Against a vanilla RAG+rerank baseline (0.674), the best graph-only configuration — $k=50$ rerank SRR with lane-gated community context — scores 0.732 (+0.058). Adding breadcrumb indexing, BM25, and ADF lifts this to 0.739 (+0.065). Ten configurations fall within the ± 0.038 statistical top band (0.700–0.739), confirming broad reproducibility of the HARE gains across k -settings and auxiliary features. The sovereign model (`gpt-oss-120b`) scores 0.600 on the equivalent configuration — below the proprietary leader but substantially above vanilla, confirming that retrieval gains transfer to open-weight deployment.

On H2 — whether stronger generators amplify retrieval gains — the cross-model comparison (Mini=0.679, Sonnet=0.643, Haiku=0.621 at k30 rerank SRR) shows Mini outperforming both Claude models. This result is confounded by prompt alignment: the SRR prompt was developed against gpt-4o-mini, and the Claude-specific suffix that suppresses attribution preambles does not fully resolve the stylistic mismatch. The cross-model HARE scores are best interpreted as lower bounds on each model’s potential under properly tuned prompts, not as a ranking of generator capability.

A frontier-model hallucination audit (§9.4) confirms that the GRB judge’s inflation is asymmetric, not additive: the vanilla baseline carries a 10% false positive rate among its highest-confidence answers; the full graph stack carries 0%. Our system is not being rewarded for hallucination — it is being accurately scored. The implication cuts both ways: the measured +0.060 GRB gain is a lower bound on the true retrieval improvement, and systems that outscore ours on the GRB leaderboard may be doing so through hallucination credit rather than genuine accuracy. The GRB leaderboard rank is not a factual accuracy rank. This problem is not resolved by switching judge model families: all frontier LLMs share a training distribution that rewards stylistic confidence and plausibility, not factual correctness, and cross-model training contamination means the “independent judge” assumption is structurally violated. The HARE-Bench typed scorer applies the only standard that is genuinely independent of the systems being evaluated, which is why the two benchmarks converge at ~74% for our system rather than diverging.

Generator-agnostic result. gpt-oss-120b (Apache 2.0 open-weight, 120B parameters) scores 0.708 on GRB — within the ± 0.014 noise band of the proprietary leader (0.712) on the same laned60+community+k30+SRR configuration, with no prompt changes or model-specific tuning. This is not a claim about open-weight models in general. It is a specific finding: the NLP-primitive retrieval stack transfers to sovereign deployment without accuracy penalty. The retrieval layer is model-agnostic by construction; the generator’s role is to produce `{answer, key_claims, evidence_used}` JSON from a fixed prompt, and gpt-oss-120b’s instruction-following fidelity is sufficient for that task.

Score stability and the performance plateau. GRB’s top seven configurations cluster within 0.011 (0.701–0.712). HARE’s top ten configurations span 0.039 (0.700–0.739). Both benchmarks show that once the universal feature set is active — NER + reference expansion + lane-gating + community context + SRR — incremental tuning of auxiliary features (BM25, ADF, k-size) produces noise-level variation rather than structural gains. The stack has reached a performance plateau on both benchmarks under current configurations. Further improvement will require either a qualitatively different retrieval mechanism or a benchmark that discriminates more finely within the plateau region.

The result is not that graphs are unnecessary — it is that LLM extraction is unnecessary to build a graph that works. The broader implication returns to the motivating agent: re-prompting is a symptom of retrieval incompleteness. Each re-prompting cycle is the agent reconstructing, at inference time, the graph structure that a better retrieval layer would have pre-computed. Cheap NLP primitives are sufficient to pre-compute it —

and cheap enough to keep current.

Architectural convergence. The design choices in this paper — typed answers, delegation boundaries, deterministic contracts between probabilistic components — were not selected as novelties. They fall out as necessities once the retrieval layer is understood as a component in an agentic stack rather than a standalone system. Single-shot LLM producing end-to-end prose was never going to be the architecture; it was a starting point. The field is converging toward specialization and contracts whether it has named it yet or not: each component does one thing well and presents a typed interface to the next. The probabilistic nature of LLMs is an asset at every layer, but the interface between layers is deterministic. The typed scorer, the SRR delegation boundary, the namespace filter, the sovereign deployment finding — these are not independent features. They are instantiations of the same principle. This system is not a prediction of where agentic AI is headed. It is a description of where it already is.

On the role of human judgment. A language model can propose an interface — it has been trained on sufficient API design, protocol specifications, and architecture documents to generate plausible contracts. What it cannot do is verify that a proposed contract is correct for a specific system, a specific failure mode, or an upstream consumer that does not yet exist in its training data. QC requires ground truth, and ground truth requires someone who knows what the system is actually supposed to do. In novel work, that is only the person who conceived it. The deeper point is about creativity: a frontier model’s generative range is recombination weighted by prior human expression. Genuinely novel architectural patterns — the ones that have not been named yet, that fall out of problems no one has previously framed — are not in the weights. The model can help express and refine an idea once it exists. It does not originate one. This is not a limitation that scale resolves; it is a structural property of how the weights were formed. The human contribution to systems like this one is not prompt engineering — it is the architectural conception that gave the prompts something real to work with.

Future work. Agentic query planning with retry loops may allow smaller models to recover the performance gaps observed on cross-document join tasks. Where a single-shot retrieval pass fails to surface the bridging evidence, a planner that re-issues targeted sub-queries — informed by what was retrieved in the previous pass — could close the gap without requiring a stronger generator. Whether this recovery is retrieval-architecture-dependent (i.e., graph-augmented retrieval gains more from retry loops than vanilla retrieval) is an open question. A GLEIF ablation (bridging documents present vs. absent) is planned to isolate whether CE/MDJ score differences reflect corpus construction or retrieval architecture; if GLEIF records materially lift CE scores, the implication is that entity alias coverage in the corpus is the primary driver of cross-entity question performance — not retrieval strategy.

Appendix A: Full Grid Results

45 GraphRAG-Bench grid runs (n=300 questions each), sorted descending by All score. Scores are answer_correctness per question type (FR = Fact Retrieval, CR = Complex

Reasoning, CS = Contextual Summarize, CG = Creative Generation). All = unweighted mean across four types.

Run	n	FR	CR	CS	CG	All
ner_ref_rerank_laned_community_k10_grid	300	0.704	0.639	0.734	0.710	0.697
ner_ref_rerank_cluster_community_k10_grid	300	0.708	0.637	0.698	0.693	0.684
ner_ref_rerank_laned55_community_k10_grid	300	0.700	0.658	0.741	0.636	0.684
ner_ref_rerank_laned_pruned_k10_grid	300	0.710	0.619	0.685	0.714	0.682
bc_ner_ref_rerank_laned_community_k10_grid	300	0.696	0.641	0.684	0.668	0.672
ner_ref_rerank_laned_community_pruned_k10_grid	300	0.685	0.623	0.740	0.639	0.672
bc_ner_ref_rerank_laned_community_pruned_k10_grid	300	0.709	0.620	0.724	0.632	0.671
ner_ref_rerank_cluster_laned_community_pruned_k10_grid	300	0.723	0.631	0.743	0.587	0.671
bc_ner_ref_rerank_laned_community_pruned_k10_grid	300	0.697	0.640	0.704	0.642	0.671
ner_ref_rerank_laned_community_alpha0_grid	300	0.683	0.612	0.678	0.702	0.669
ner_ref_rerank_laned_community_k20_grid	300	0.696	0.590	0.726	0.660	0.668
ner_ref_rerank_laned_community_pruned_k20_grid	300	0.687	0.678	0.734	0.573	0.668
ner_ref_rerank_laned_community_grid	300	0.694	0.594	0.734	0.640	0.665
ner_ref_rerank_laned_k10_grid	300	0.708	0.628	0.715	0.609	0.665
ner_ref_rerank_laned_community_k25_grid	300	0.686	0.619	0.698	0.652	0.664
ner_ref_rerank_laned60_community_k10_grid	300	0.702	0.625	0.653	0.675	0.664
rerank_k10_grid	300	0.707	0.643	0.743	0.558	0.663
rerank_laned_community_k10_grid	300	0.710	0.651	0.698	0.593	0.663
semantic_boundary_rerank_grid	300	0.697	0.642	0.723	0.589	0.663
ner_ref_rerank_laned_community_pruned_k25_grid	300	0.702	0.656	0.690	0.595	0.661
ner_ref_rerank_cluster_community_grid	300	0.699	0.585	0.677	0.682	0.661
ner_ref_rerank_laned_community_k15_grid	300	0.684	0.613	0.739	0.605	0.660
ner_ref_rerank_laned60_grid	300	0.692	0.639	0.714	0.596	0.660
ner_ref_rerank_laned_community_k7_grid	300	0.698	0.631	0.723	0.588	0.660
bc_ner_rerank_grid	300	0.669	0.609	0.751	0.608	0.659
ner_ref_rerank_laned_community_pruned_k15_grid	300	0.676	0.636	0.709	0.611	0.658
bc_ner_ref_rerank_laned_community_grid	300	0.682	0.644	0.686	0.612	0.656
ner_ref_rerank_pruned_laned_community_grid	300	0.671	0.616	0.718	0.614	0.655
ner_ref_rerank_laned_grid	300	0.683	0.605	0.701	0.626	0.654
ner_ref_rerank_community_k10_grid	300	0.692	0.640	0.699	0.584	0.654
bc_ner_ref_rerank_laned_k10_grid	300	0.705	0.664	0.663	0.581	0.653
ner_ref_rerank_laned_community65_k10_grid	300	0.714	0.601	0.721	0.574	0.652
ner_ref_rerank_community_grid	300	0.679	0.625	0.722	0.583	0.652
ref_laned_community_k10_norerank_grid	300	0.660	0.587	0.689	0.671	0.652
ner_rerank_laned_community_k10_grid	300	0.704	0.623	0.689	0.588	0.651
ner_ref_rerank_cluster_laned_grid	300	0.704	0.604	0.723	0.565	0.649
ner_ref_rerank_pruned_grid	300	0.710	0.615	0.700	0.529	0.639
ner_ref_rerank_grid	300	0.697	0.580	0.694	0.582	0.638
ner_ref_rerank_laned_bctx_grid	300	0.706	0.611	0.666	0.564	0.637
ner_ref_rerank_laned55_community65_grid	300	0.677	0.619	0.665	0.583	0.636
ner_ref_rerank_laned_pruned_grid	300	0.671	0.615	0.677	0.580	0.636
ner_ref_rerank_v2_grid	300	0.712	0.604	0.681	0.536	0.633
vanilla_256_rerank_grid	300	0.674	0.591	0.683	0.581	0.632
ner_rerank_grid	300	0.665	0.628	0.678	0.548	0.630

Run	n	FR	CR	CS	CG	All
ner_ref_rerank_laned_community_bctx_grid	300	0.696	0.610	0.672	0.536	0.628

Vanilla baseline: `vanilla_256_rerank_grid` (All = 0.632). Best graph configuration: `ner_ref_rerank_laned_community_k10_grid` (All = 0.697, +0.065 over vanilla). The top 16 configurations all exceed 0.660; the top 4 are within 0.013 of each other, confirming that configuration sensitivity is low once core graph features are active.

Appendix B: Statistical Significance

Bootstrap 95% CIs (n=10,000 resamples, paired by question ID, seed=42) for key pairwise comparisons. GRB scores use `answer_correctness`; HARE-Bench scores use `typed_score`. All comparisons are on full-corpus runs (GRB: n=4,071; HARE-Bench: n=500).

Comparison	Baseline	System	Δ	95% CI	$p(\Delta \leq 0)$
GRB: vanilla+rerank vs graph+SRR	0.656	0.706	+0.050	[+0.041, +0.059]	<0.001
GRB: graph (no SRR) vs graph+SRR	0.671	0.706	+0.035	[+0.027, +0.043]	<0.001
HARE-Bench: Mini vs Haiku at k30+rerank+SRR	0.675 (Mini)	0.618 (Haiku)	-0.057	[-0.087, -0.026]	<0.001
HARE-Bench: vanilla+rerank vs graph+SRR (Mini)	0.669	0.675	+0.006	[-0.022, +0.033]	0.340

Interpretation.

GRB retrieval gains are statistically reliable at full-corpus scale. The +0.050 vanilla→graph+SRR gain (CI entirely above zero) and +0.035 SRR-isolated gain are both significant at $p < 0.001$. These confirm the main GRB conclusions in §6 are not sampling artifacts.

The Haiku deficit on HARE-Bench (-0.057, CI: [-0.087, -0.026]) is statistically significant. The capability gap is real, not a measurement fluctuation; see §7.4 and §8.4 for mechanism analysis.

The HARE-Bench vanilla→graph+SRR gain (+0.006, CI: [-0.022, +0.033]) is not statistically significant ($p=0.34$). HARE-Bench gains are concentrated in the post-canonical calibration run (§8.7), where the absolute improvement over vanilla is +0.046–0.069 depending on configuration. The non-significance of the paired test on the full 500-question set reflects the difficulty of the benchmark and the tighter question-level variance on a heterogeneous corpus. The directional gain holds; the point estimate alone does not clear the significance threshold.

The §6.5 note (“*Treat the statistical tie observations below as directional*”) applies to within-band GRB grid results (n=300 per configuration, $MDD \approx \pm 0.015$). Full-corpus

GRB results (n=4,071) have substantially narrower CIs and the main comparisons are significant.

Appendix C: Experimental Run Reference

Script	Runs	Status
run_bc_grid_missing.sh	6	☐ complete
run_ablation_structural.sh	2	☐ complete
run_grid_gaps.sh	2	☐ complete
run_new_ideas.sh	6	☐ complete
run_paper_validation.sh	4	☐ complete
run_extended_grid.sh	8	☐ complete
run_full_all.sh	GraphRAG-Bench full-corpus: Mini + Haiku + sovereign (gpt-oss-120b)	☐ complete
run_fang2026.sh	HARE-Bench: 43-run full matrix (Mini, Haiku, Sonnet, gpt-oss-120b; k30/k50; BM25/BC/ADF variants)	☐ complete

Appendix D: Run Name Key

Run names encode the retrieval configuration as a _-delimited sequence of feature tokens. All graph-augmented runs use semantic boundary chunking (1100–2200 tokens). The mapping:

Token	Meaning
vanilla_256	Naive 256-token fixed-size chunking. No semantic boundaries, no graph features.
ner_ref	NER entity-ref-expansion enabled (P1)
rerank	Cross-encoder reranking (BAAI/bge-reranker-large)
laned	Lane filtering at default $\text{sim} \geq 0.45$ threshold (see also laned55, laned60)
laned55 / laned60	Lane filtering at $\text{sim} \geq 0.55/0.60$
community	Community context injection (P2); coherence filter default 0.50
community60	Community context with $\text{coherence} \geq 0.60$
pruned	Redundancy pruning at $\text{cosine} \geq 0.92$

Token	Meaning
cluster	Cluster-based topological expansion (replaces lane filtering for heterogeneous corpora)
kN	Retrieval depth top-k=N (default k=5 if omitted)
global_search	Global search mode (community-level summarization retrieval)
graph_first	Graph-first search mode (entity traversal before dense retrieval)
haiku	Generator: claude-haiku-4-5-20251001 via Anthropic
llama8b	Generator: Llama-3.1-8B-Instruct-Turbo via Together AI
srr	Evidence-compliance reprompt — generator produces {answer, key_claims, evidence_used} JSON; fires once when evidence_used=[]; no additional retrieval; strictly additive
full	Full-corpus run (4,072 questions)
grid	Grid sweep run (300 stratified questions)

Examples: -ner_ref_rerank_laned_community_pruned_k20_full = semantic boundary chunking + entity-ref-expansion + reranking + lane filter (sim $\geq$ 0.45) + community context + redundancy pruning + k=20, full corpus -ner_ref_laned60_community_k30_haiku_full = semantic boundary chunking + entity-ref-expansion + lane filter (sim $\geq$ 0.60) + community + k=30, Haiku generator, full corpus -fang_ner_ref_cluster_community_k10_haiku = HARE-Bench corpus + entity-ref + cluster expansion + community context + k=10, Haiku generator